

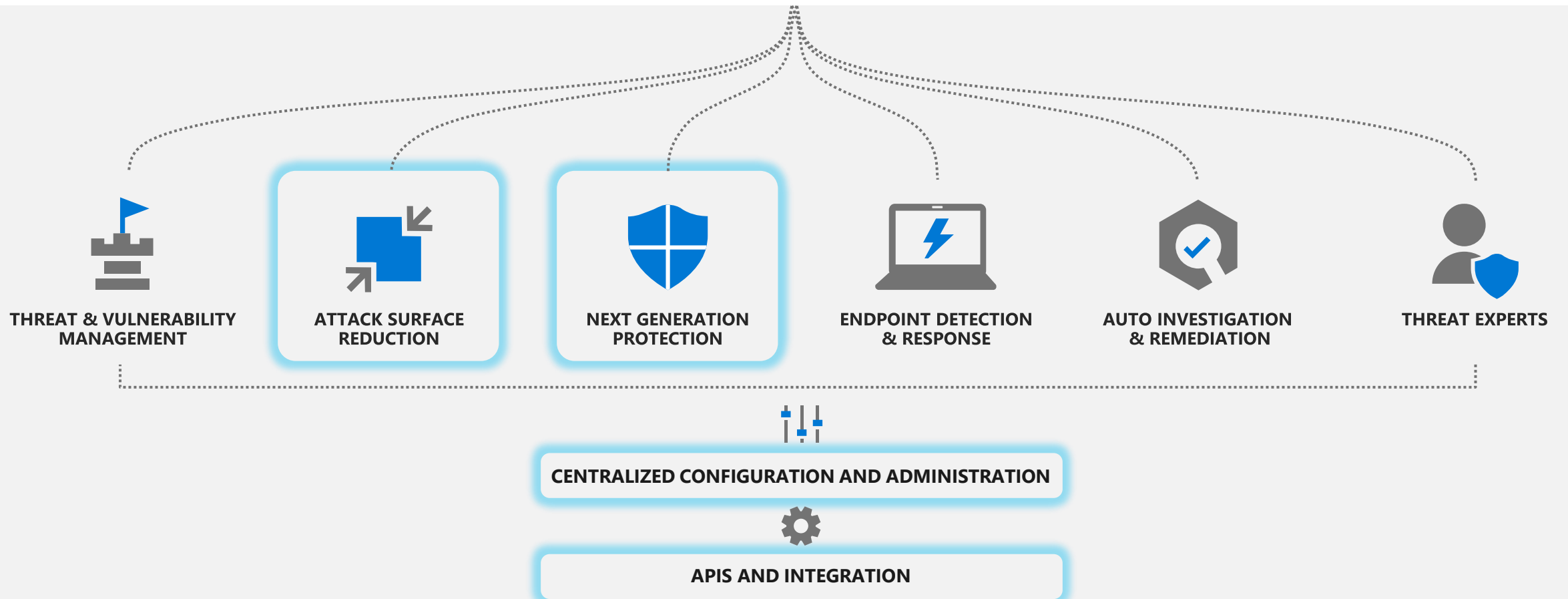
NG Endpoint Protection





Next Generation Endpoint Protection

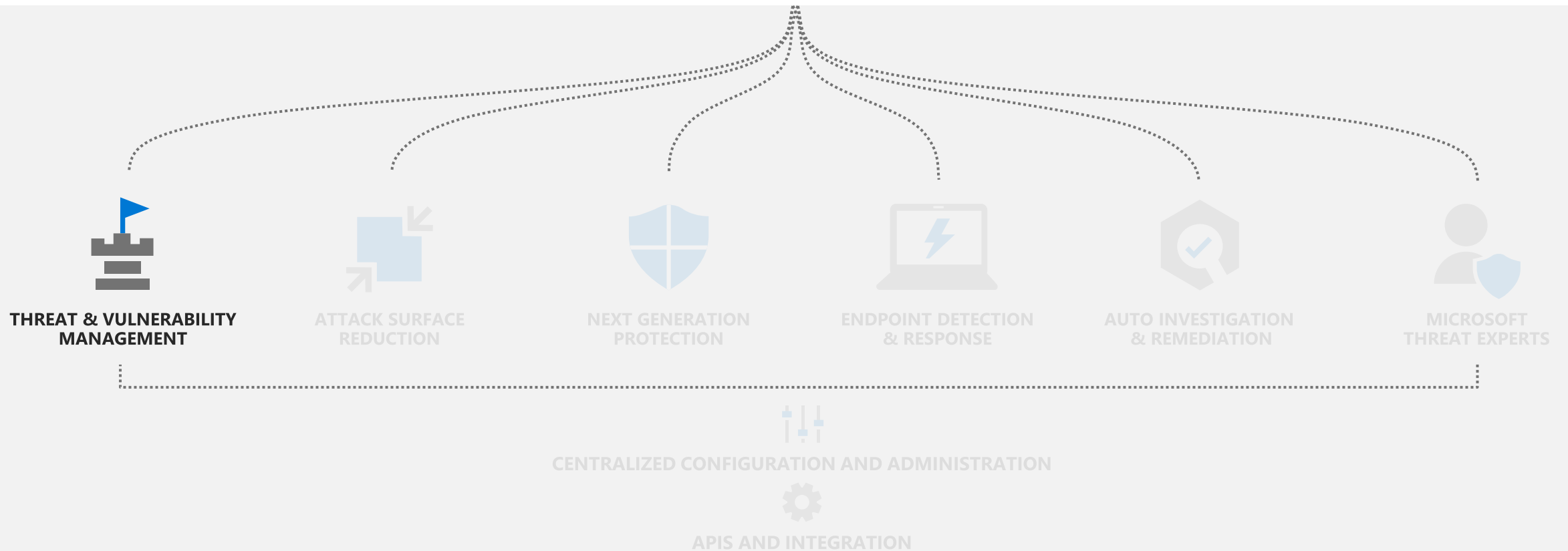
Threats are no match.





Next Generation Endpoint Protection

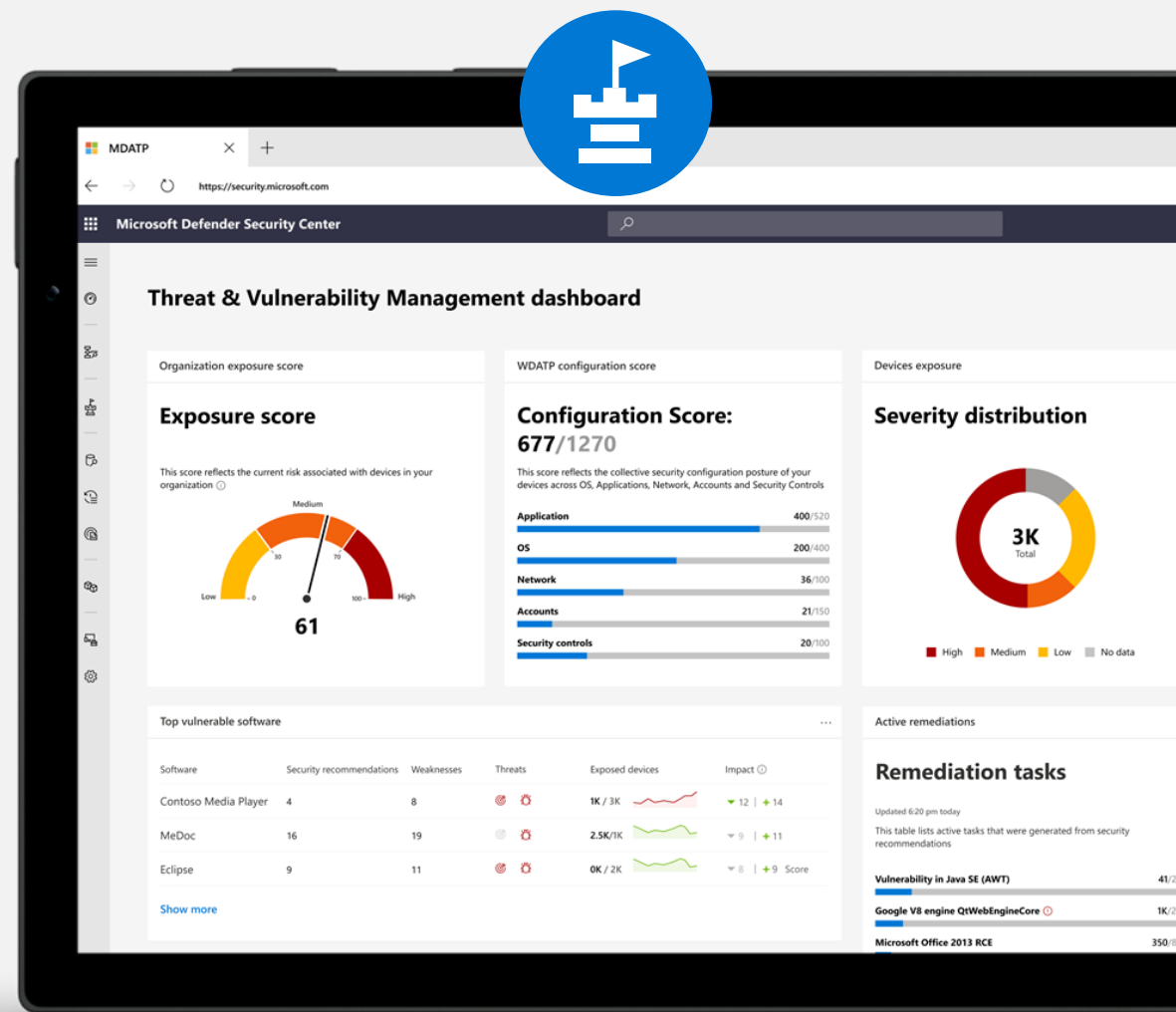
Threats are no match.



Threat & Vulnerability Management

A risk-based approach to mature your vulnerability management program

- 1  Continuous real-time discovery
- 2  Context-aware prioritization
- 3  Built-in end-to-end remediation process



1



Continuous Discovery

Extensive vulnerability assessment across the entire stack

Easiest to exploit



Application extension vulnerabilities

Application-specific vulnerabilities that relate to component within the application.
For example: Grammarly Chrome Extension (CVE-2018-6654)



Application run-time libraries vulnerabilities

Reside in a run-time libraries which is loaded by an application (dependency).
For example: Electron JS framework vulnerability (CVE-2018-1000136)



Application vulnerabilities (1st and 3rd party)

Discovered and exploited on a daily basis.
For example: 7-zip code execution (CVE-2018-10115)



OS kernel vulnerabilities

Becoming more and more popular in recent years due to OS exploit mitigation controls.
For example: Win32 elevation of privilege (CVE-2018-8233)



Hardware vulnerabilities (firmware)

Extremely hard to exploit, but can affect the root trust of the system.
For example: Spectre/Meltdown vulnerabilities (CVE-2017-5715)

Hardest to discover

1



Continuous Discovery

Broad secure configuration assessment



Operation system misconfiguration

- File Share Analysis
- Security Stack configuration
- OS baseline



Application misconfiguration

- Least-privilege principle
- Client/Server/Web application analysis
- SSL/TLS Certificate assessment



Account misconfiguration

- Password Policy
- Permission Analysis



Network misconfiguration

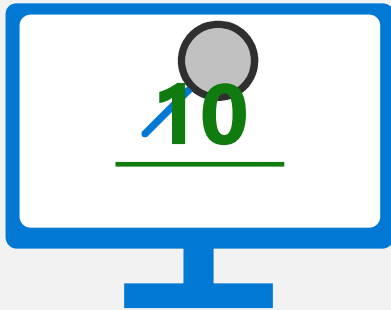
- Open ports analysis
- Network services analysis

2



Threat & Business Prioritization ("TLV")

Helping customers focus on the right things at the right time



Threat Landscape

Vulnerability characteristics (CVSS score, days vulnerable)
Exploit characteristics (public exploit & difficulty, bundle)
EDR security alerts (Active alerts, breach history)
Threat analytics (live campaigns, threat actors)



Breach Likelihood

Current security posture
Internet facing
Exploit attempts in the org



Business Value

HVA analysis (WIP, HVU, critical process)
Run-time & Dependency analysis

3



Automated Compensation

Bridging between the IT and Security admins

Game changing bridge between IT and Security teams

1-click remediation requests via Intune/SCCM

Automated task monitoring via run-time analysis

Tracking Mean-time-to-mitigate KPIs

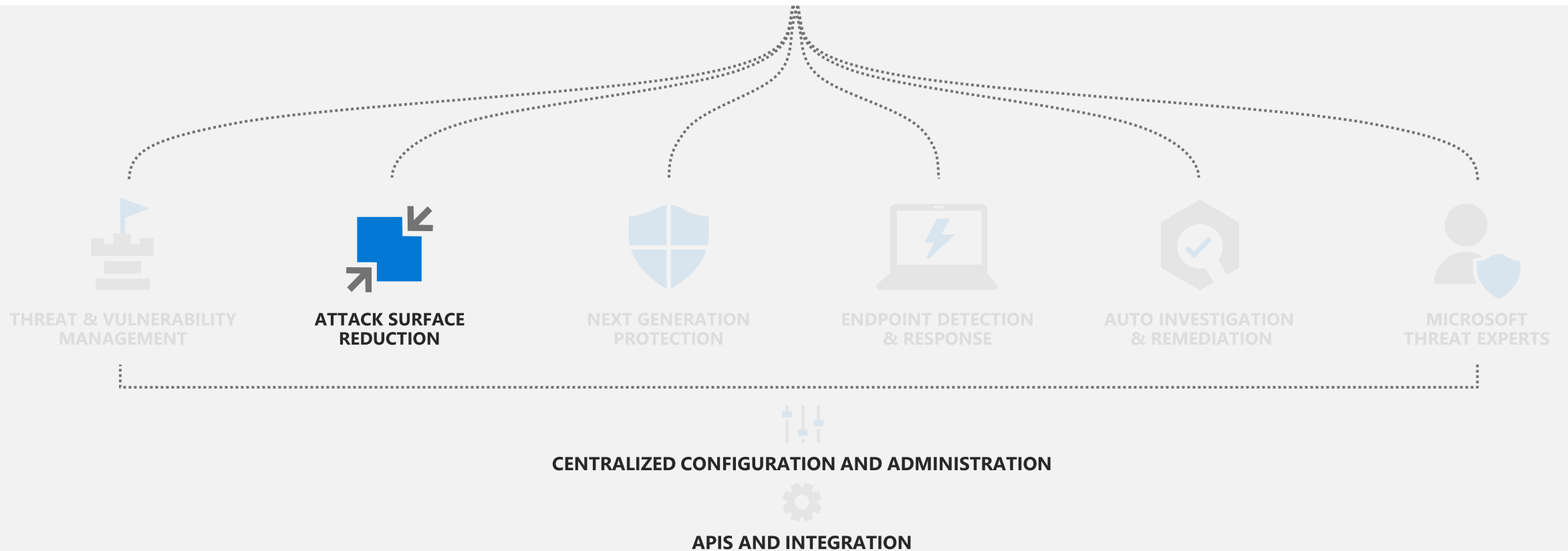
Rich exception experience to mitigate/accept risk

Ticket management integration (Intune, Planner, Service Now, JIRA)



Next Generation Endpoint Protection

Threats are no match.



Attack Surface Reduction

Eliminate risks by reducing the surface area of attack



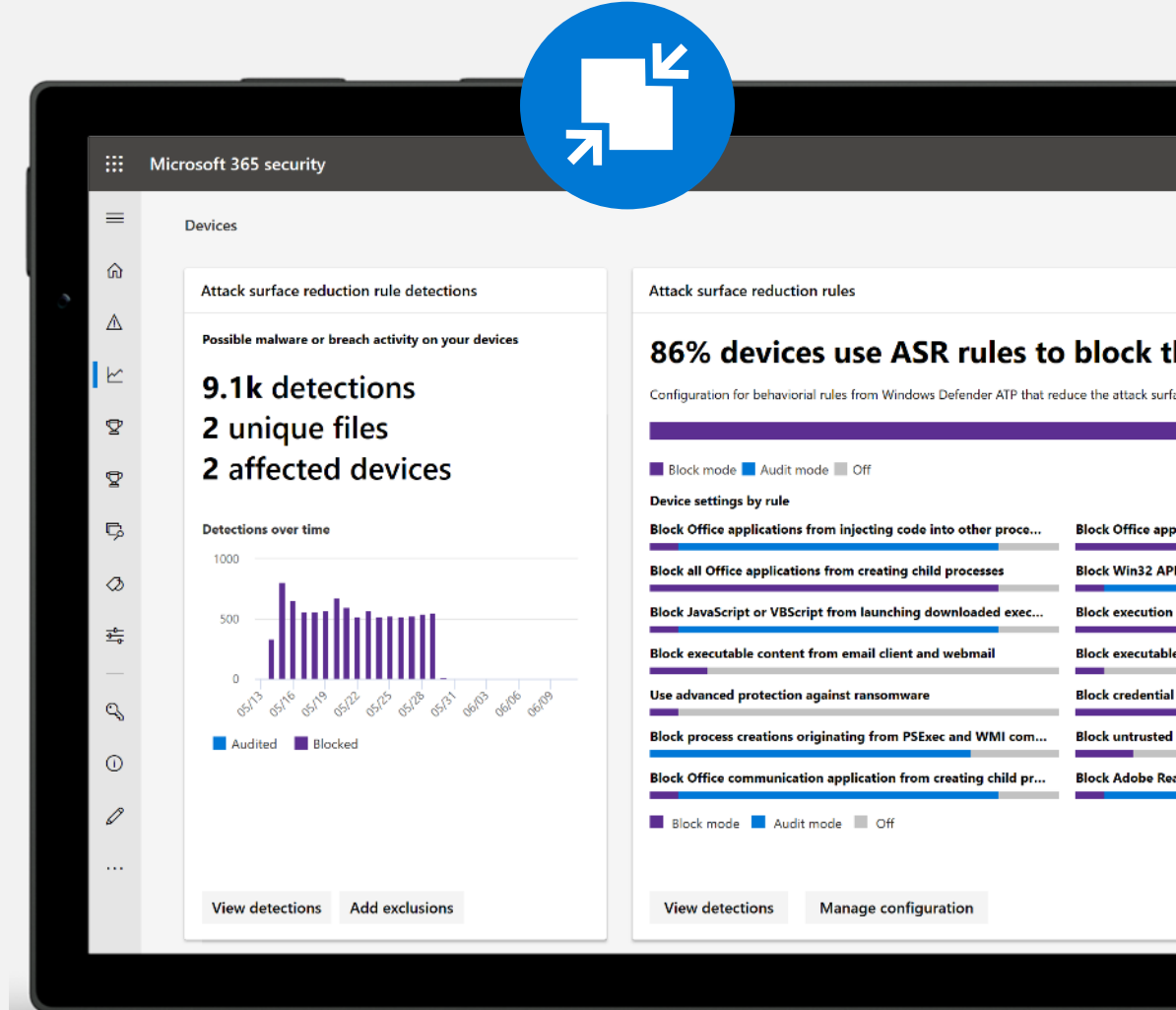
System hardening without disruption



Customization that fits your organization

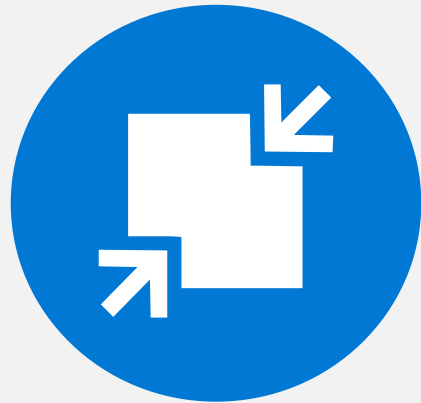


Visualize the impact and simply turn it on



Attack Surface Reduction

Resist attacks and exploitations



HW based isolation

Application control

Exploit protection

Network protection

Controlled folder access

Device control

Web protection

Ransomware protection

Isolate access to untrusted sites

Isolate access to untrusted Office files

Host intrusion prevention

Exploit mitigation

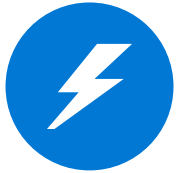
Ransomware protection for your files

Block traffic to low reputation destinations

Protect your legacy applications

Only allow trusted applications to run

Attack Surface Reduction (ASR) Rules



Minimize the attack surface

Signature-less, control entry vectors, based on cloud intelligence. Attack surface reduction (ASR) controls, such as behavior of Office macros.

Productivity apps rules

- Block Office apps from creating executable content
- Block Office apps from creating child processes
- Block Office apps from injecting code into other processes
- Block Win32 API calls from Office macros
- Block Adobe Reader from creating child processes

Email rule

- Block executable content from email client and webmail
- Block only Office communication applications from creating child processes

Script rules

- Block obfuscated JS/VBS/PS/macro code
- Block JS/VBS from launching downloaded executable content

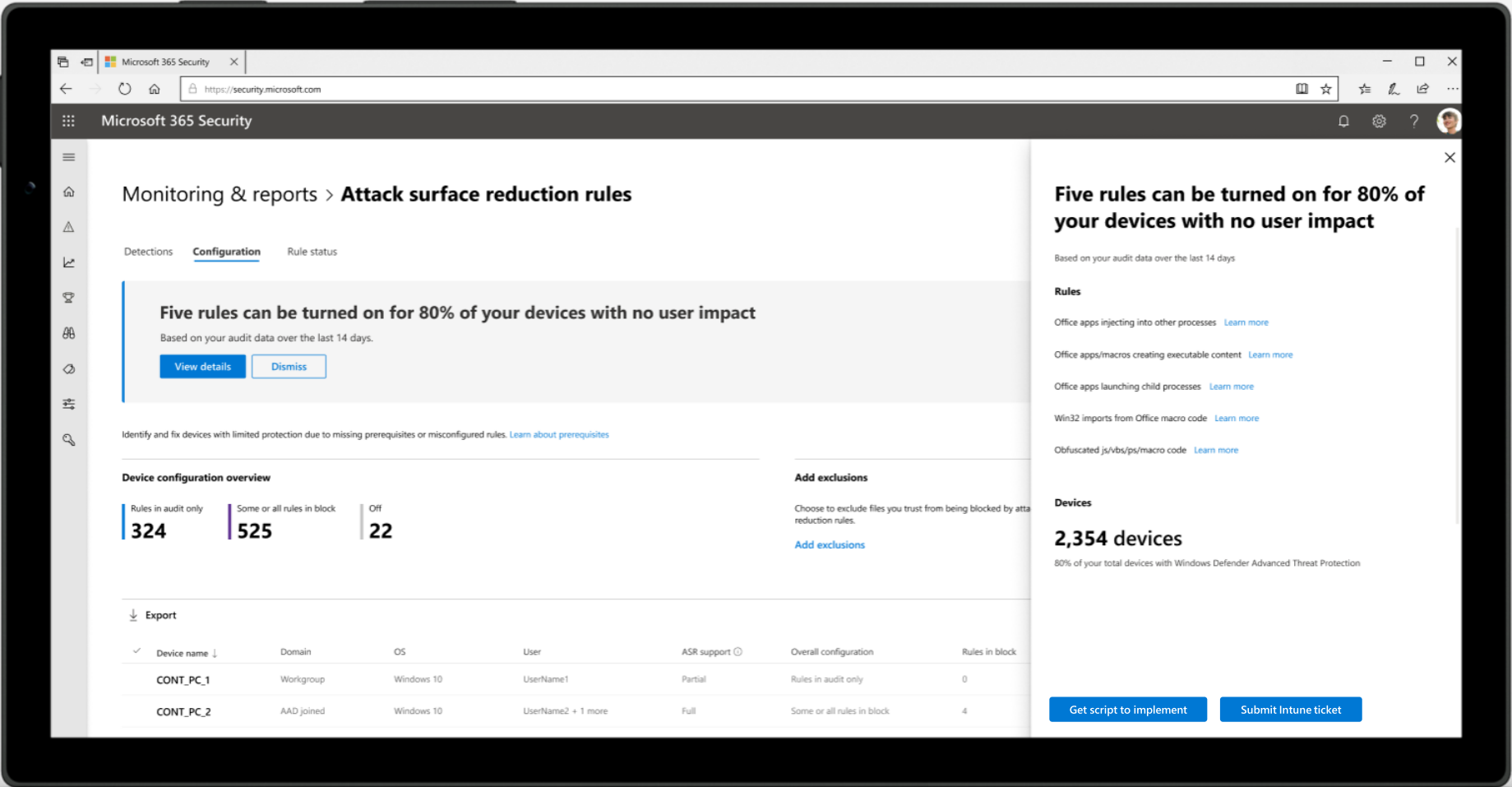
Polymorphic threats

- Block executable files from running unless they meet a prevalence (1000 machines), age (24hrs), or trusted list criteria
- Block untrusted and unsigned processes that run from USB
- Use advanced protection against ransomware

Lateral movement & credential theft

- Block process creations originating from PSEXEC and WMI commands
- Block credential stealing from the Windows local security authority subsystem (lsass.exe)
- Block persistence through WMI event subscription

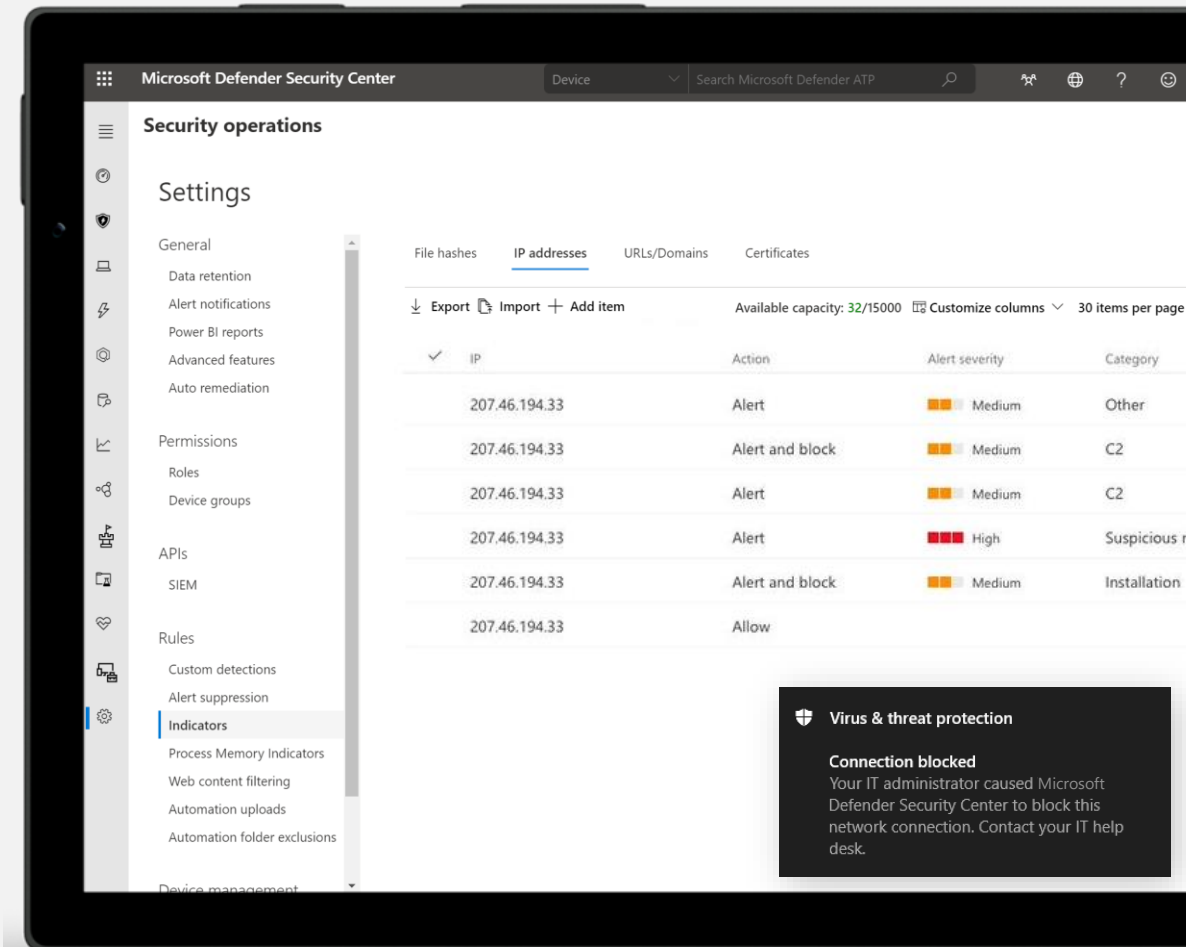
Easy button: turn on block



Network protection

Allow, audit and block

- Perimeter-less network protection (“SmartScreen in the box”) preventing users from accessing malicious or suspicious network destinations, **using any app on the device and not just Microsoft Edge.**
- Customers can add their own TI in additional to trusting our rich reputation database.



Web Threat Alerts

[Alerts](#) > [Suspicious connection blocked by network pro...](#)

**Suspicious connection blocked by network protection**
This alert is part of incident (76)
[Actions](#) ▾
Severity: Informational
Category: [Command And Control](#)
Detection source: EDR
Detection technology: Behavioral, Network

 Automated investigation is not applicable to alert type 

Alert context
 [minint scps](#)
 [minint](#)    
First activity: 07.29.2019 | 16:23:52
Last activity: 07.29.2019 | 16:23:52

Status
State: New
Classification: Not set
Assigned to: Not assigned

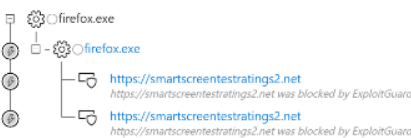
Description

Network protection prevented an attempt to connect to a malicious, compromised, or user-blocked URL, Domain, IP.

[Show more](#)

Recommended actions

1. Check the destination address. Note that highly reputable addresses might be flagged if they contain malicious content in subfolders.
2. Review the process that initiated the connection. If the process is unfamiliar and the executable not a signed system file, submit the file for deep analysis and review detailed behavioral information from the analysis results. Initiate an antivirus scan to find previously undetected malware.
3. If you've confirmed this activity to be malicious, contain and mitigate the breach. Stop suspicious processes, isolate affected machines, decommission compromised accounts or reset their passwords, block IP addresses and URLs, and install security updates.

Alert process tree

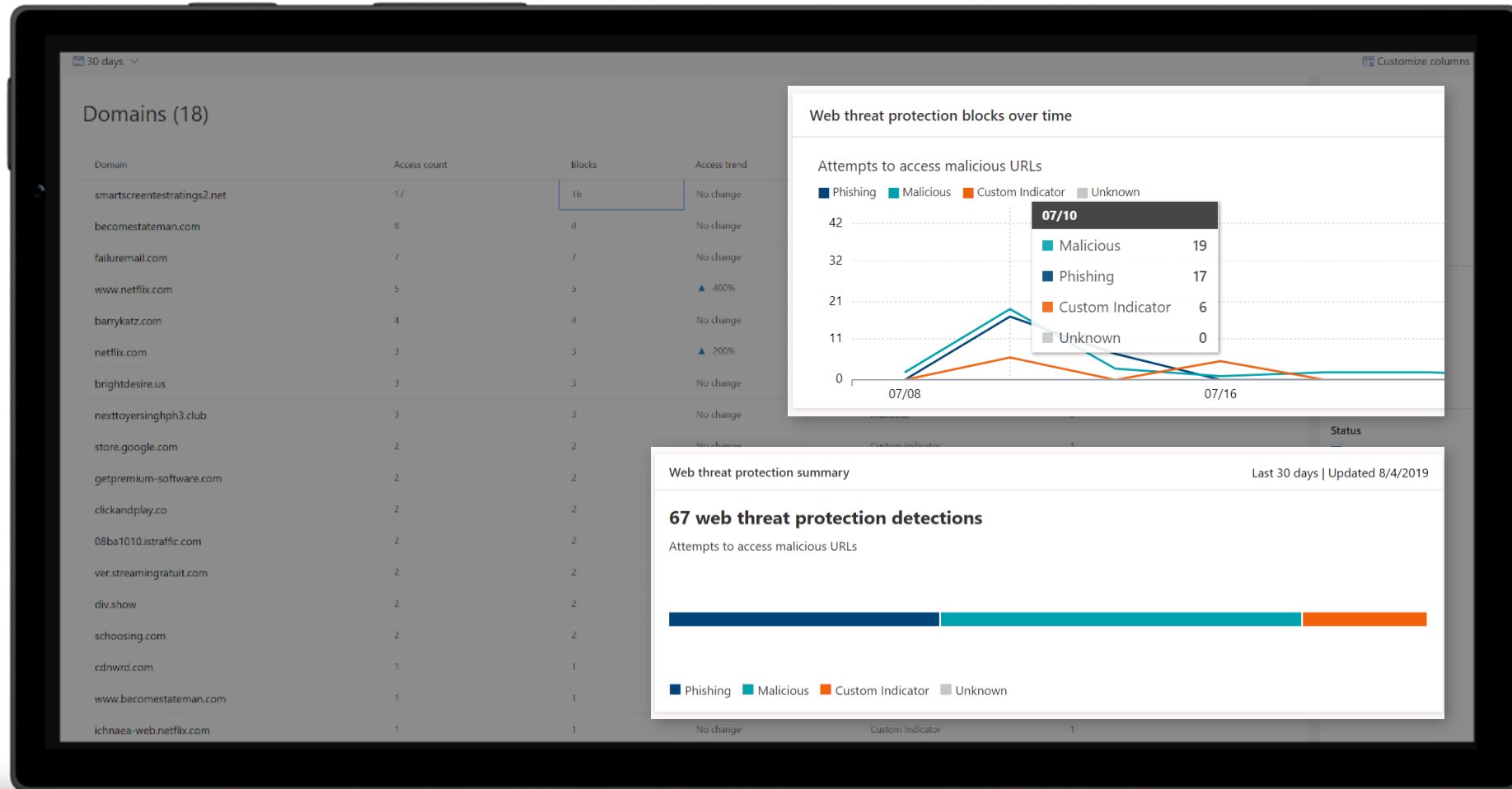
```
graph TD; A[firefox.exe] --> B[firefox.exe]; A --> C[firefox.exe]; B --> D["https://smartscreentestratings2.net  
https://smartscreentestratings2.net was blocked by ExploitGuard"]; C --> E["https://smartscreentestratings2.net  
https://smartscreentestratings2.net was blocked by ExploitGuard"];
```

 Incident graph is not available for this alert

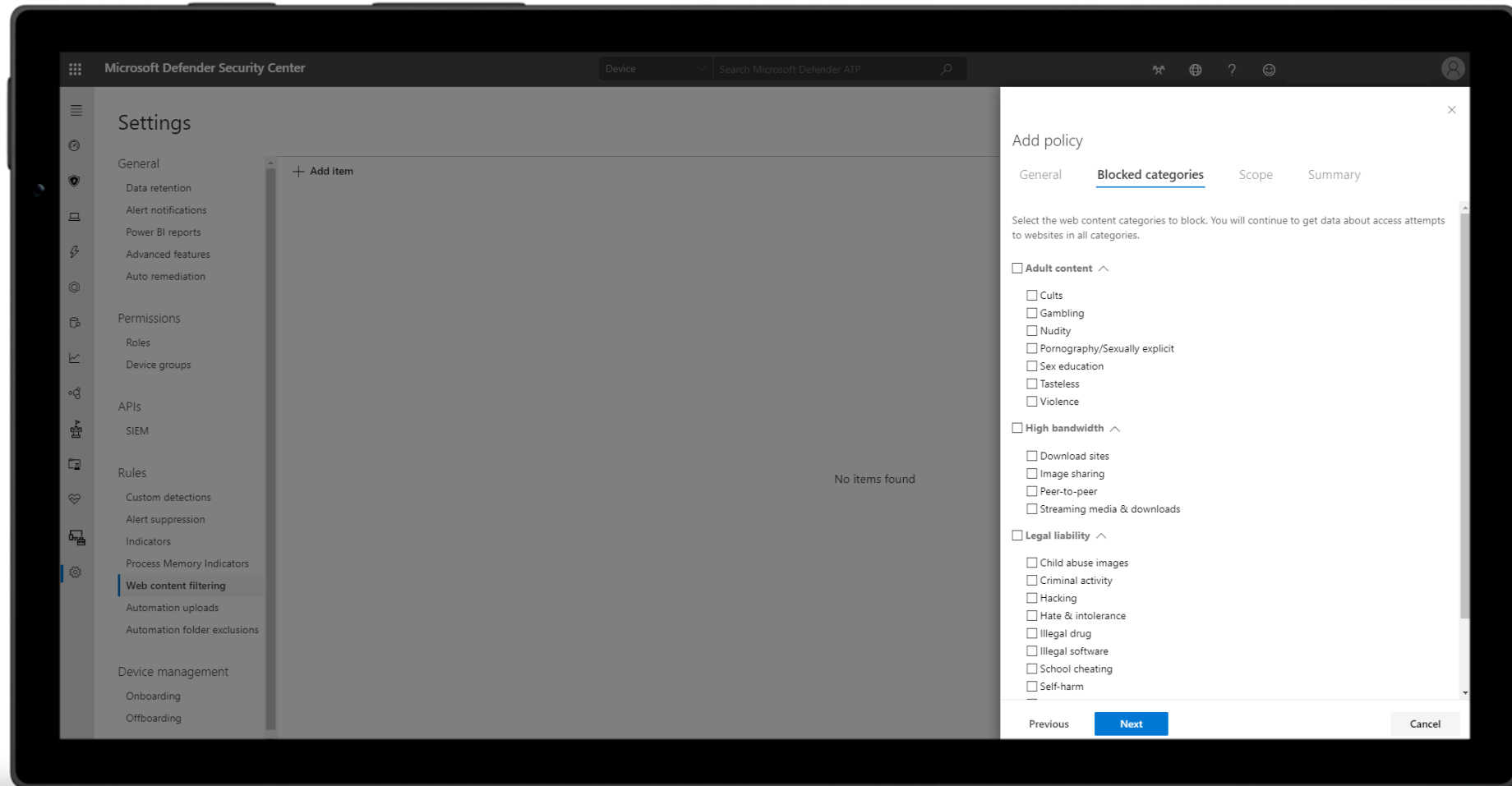
Artifact timeline

Description	First Observed	Details
-------------	----------------	---------

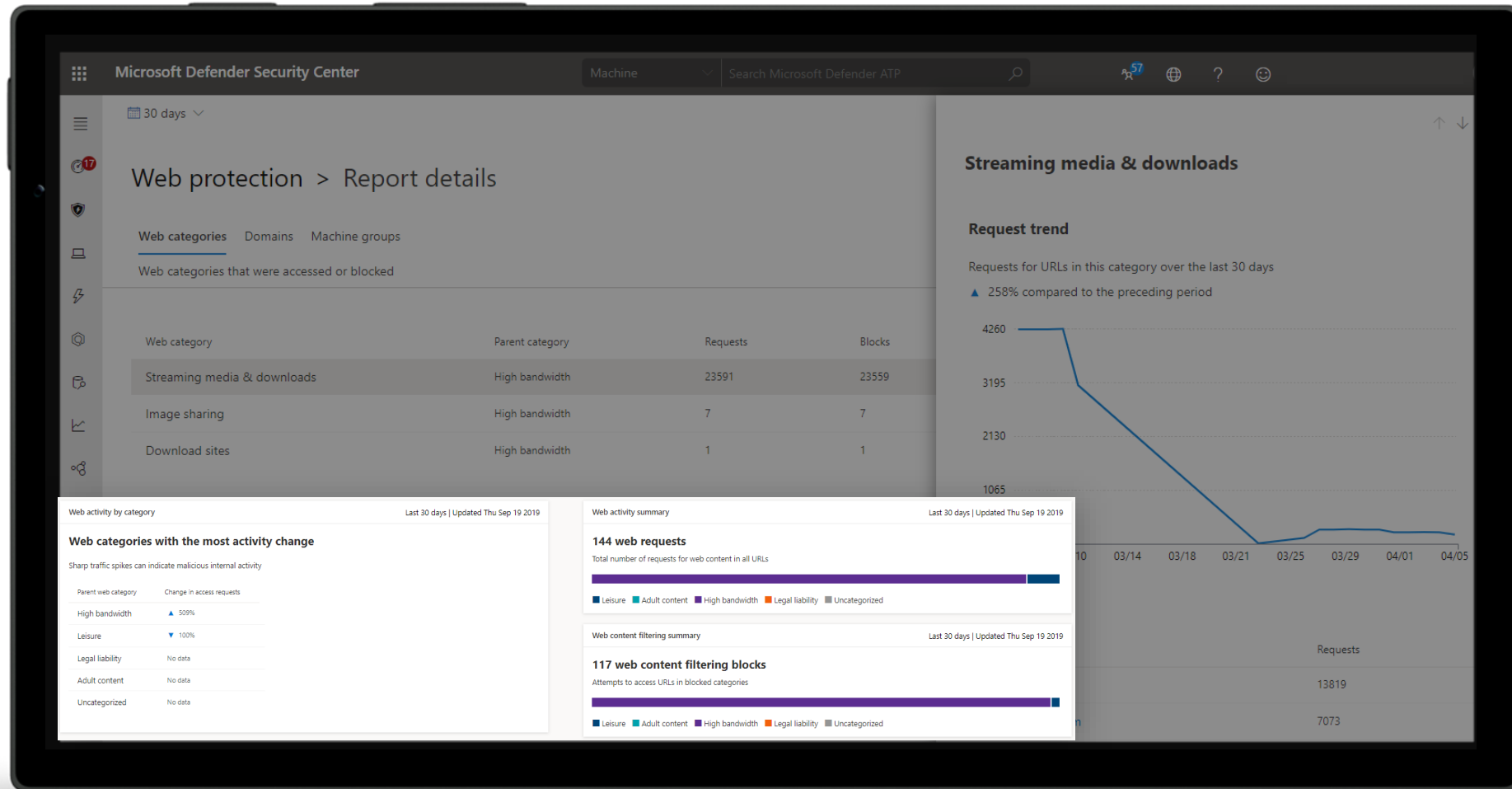
Web Threat Reports



Web content filtering configuration



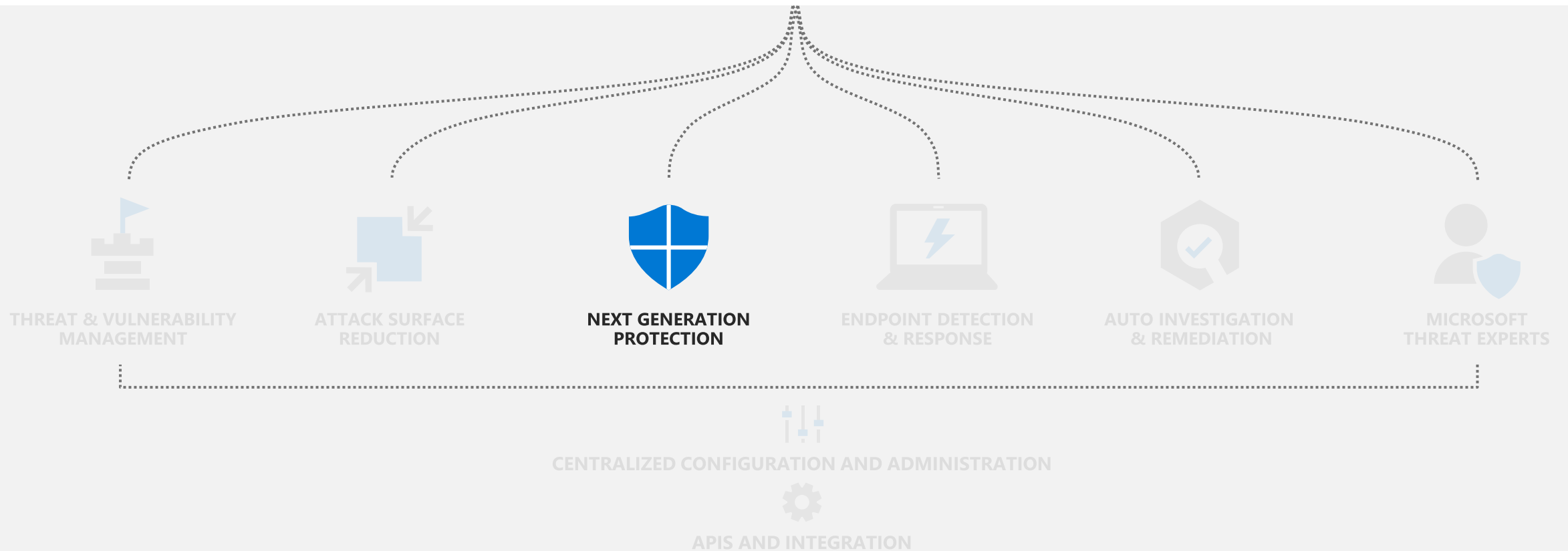
Web Content Filtering reporting





Next Generation Endpoint Protection

Threats are no match.



Static vs Dynamic

Static signatures:
focus on a file

Hashes
Strings
Emulators



Ineffective

Dynamic heuristics:
focus on *run-time behaviors*

Behavior monitoring
Memory scanning
AMSI
Command-line scanning



Effective

Next Generation Protection

Blocks and tackles sophisticated threats and malware



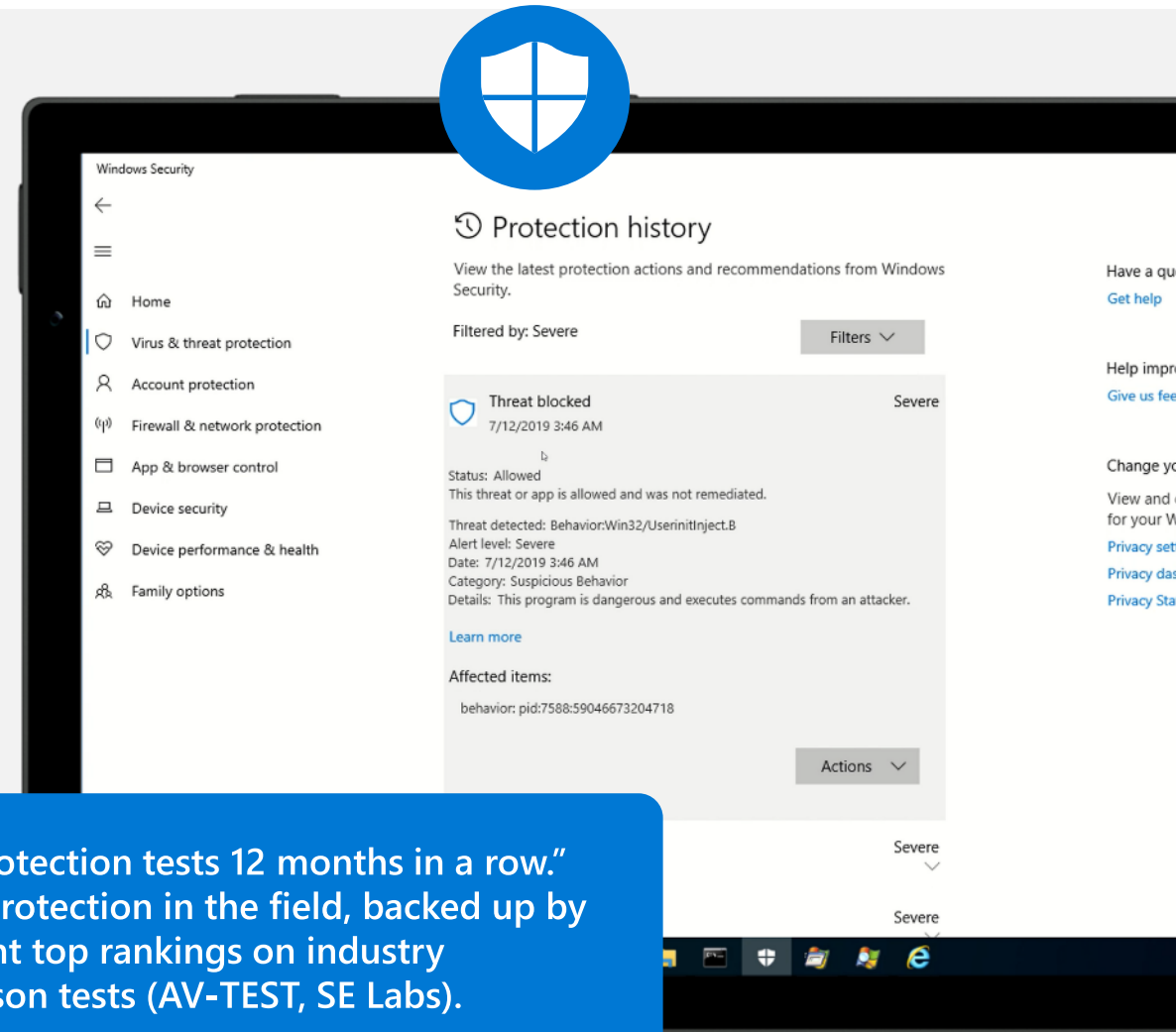
Behavioral based real-time protection



Blocks file-based and fileless malware

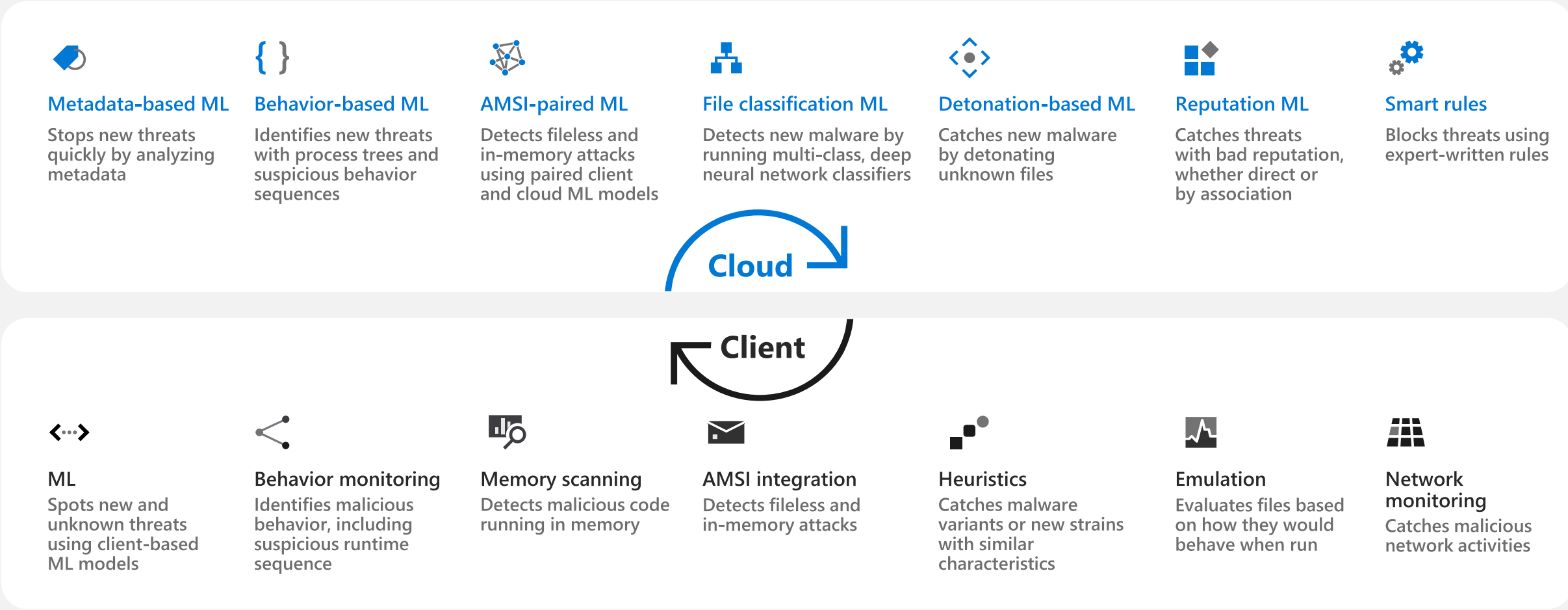


Stops malicious activity from trusted and untrusted applications



"Aced protection tests 12 months in a row."
Proven protection in the field, backed up by
consistent top rankings on industry
comparison tests (AV-TEST, SE Labs).

Next generation protection engines



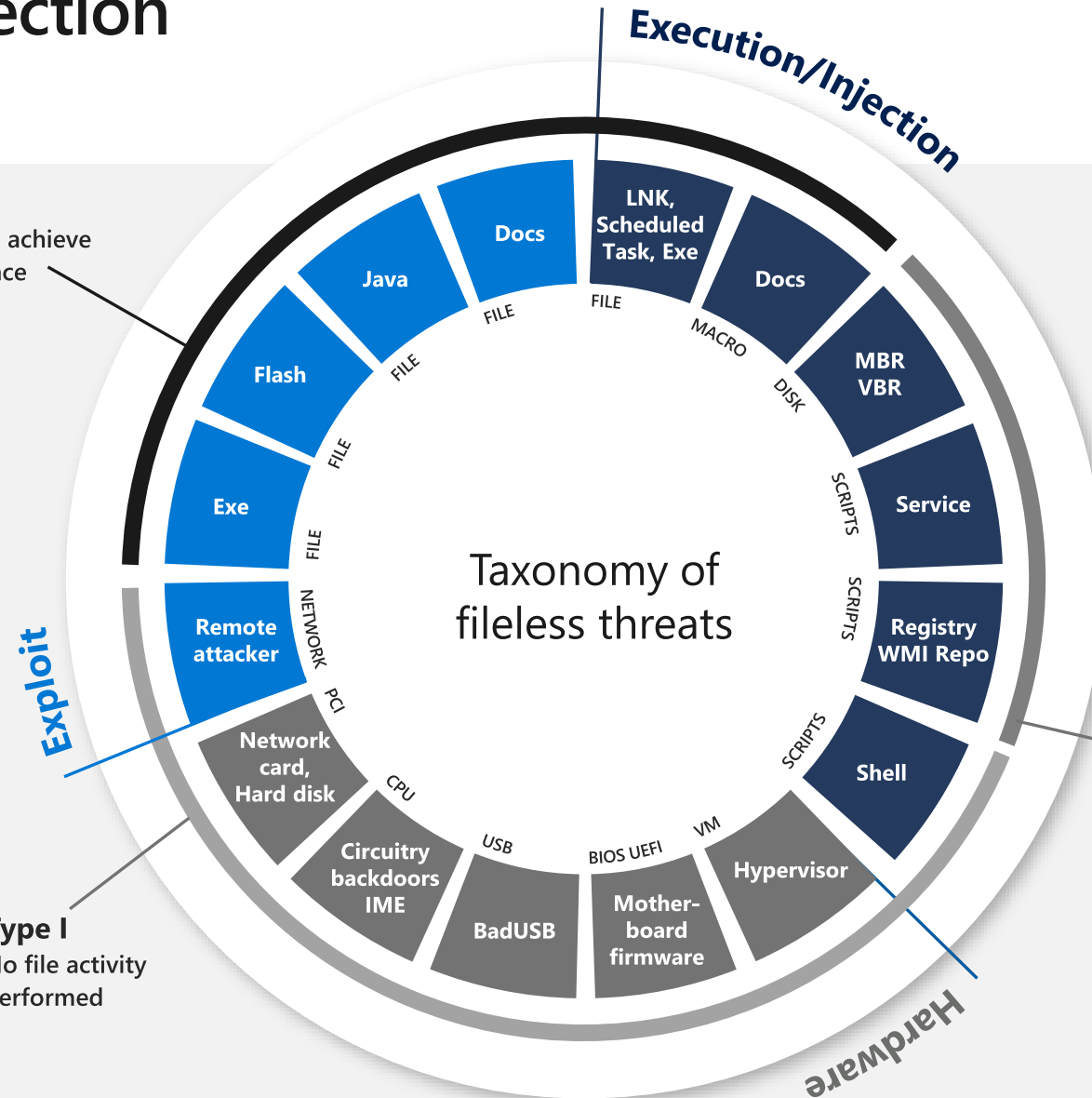
Innovations in Fileless Protection

- Dynamic and in context URL analysis to block call to malicious URL
- AMSI-paired machine learning uses pairs of client-side and cloud-side models that integrate with Antimalware Scan Interface ([AMSI](#)) to perform advanced analysis of scripting behavior
- DNS exfiltration analysis
- Deep memory analysis

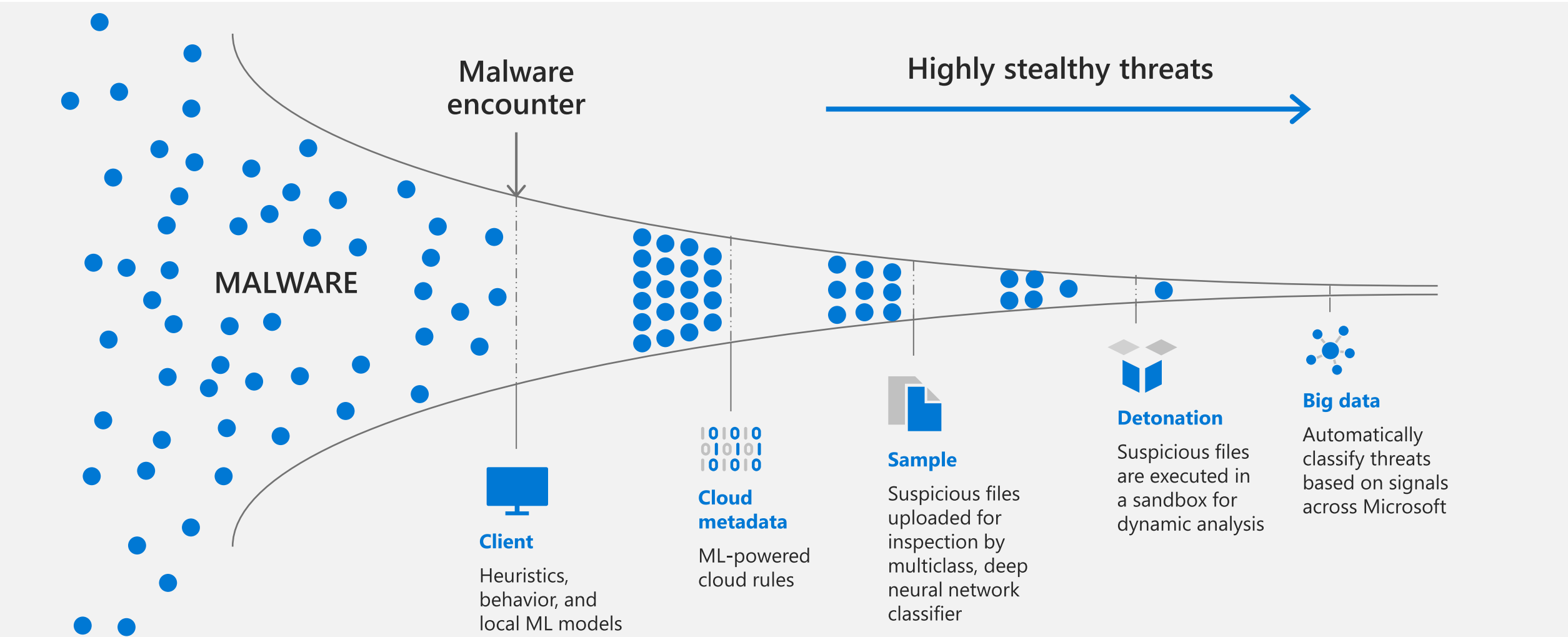
Type III
Files required to achieve
fileless persistence

Type I
No file activity
performed

Type II
No file written
on disk, but some
files used indirectly



Endpoint's NGP protection pipeline



Dynamic: behavior monitoring

Monitors activity on:



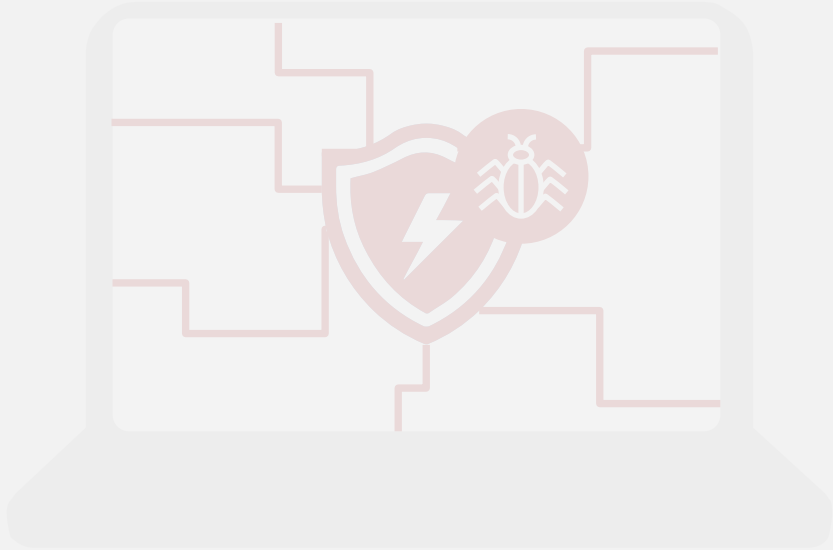
- Files
- Registry keys
- Processes
- Network (basic HTTP inspection)
- ... and few other specific activities

Heuristics can:

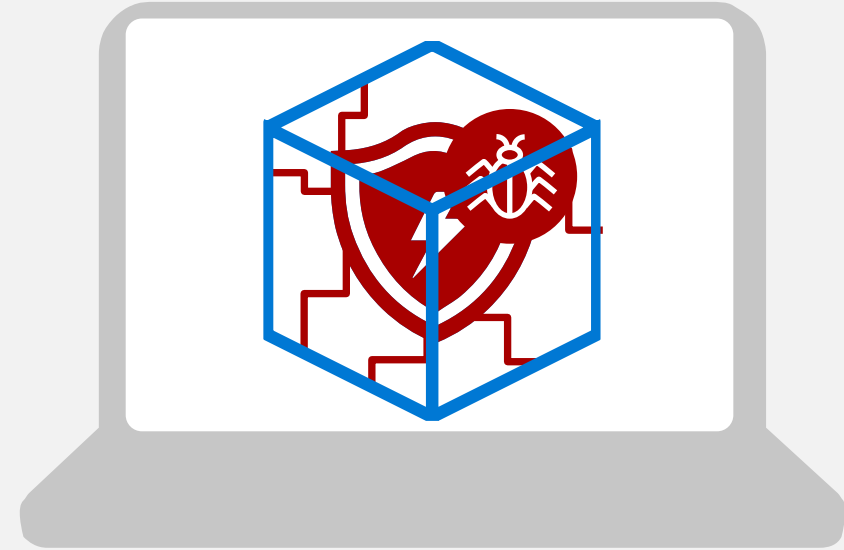


- **Detect sequences of events**
E.g. a file named "malware.exe" is created
- **Inspect event data**
E.g. an AutoRun key is created and contains "malware.exe"
- **Correlate with other static signals**
E.g. "malware.exe" has an attribute indicating it is a DotNet executable
- **Perform some basic remediation**
E.g. delete "malware.exe" if the BM event reported infection
- **Request memory scan of running processes**

Sandboxing of the antivirus engine



Then



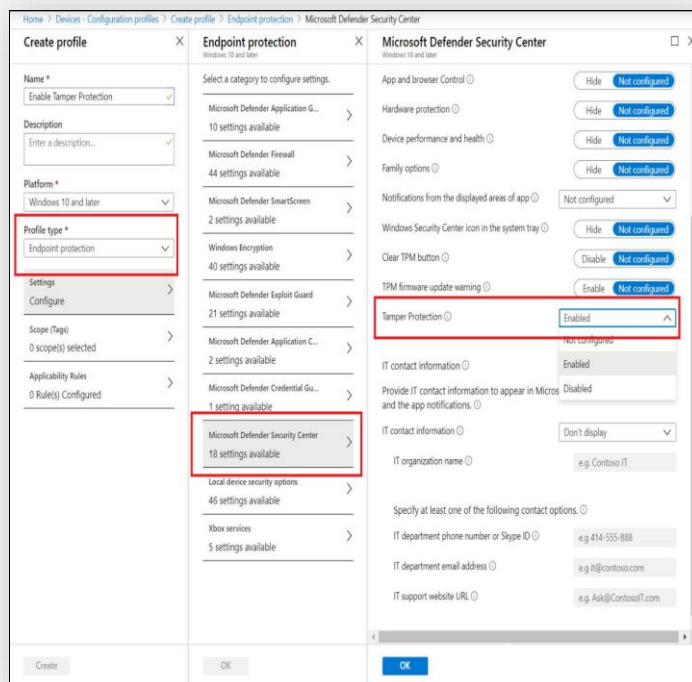
Now



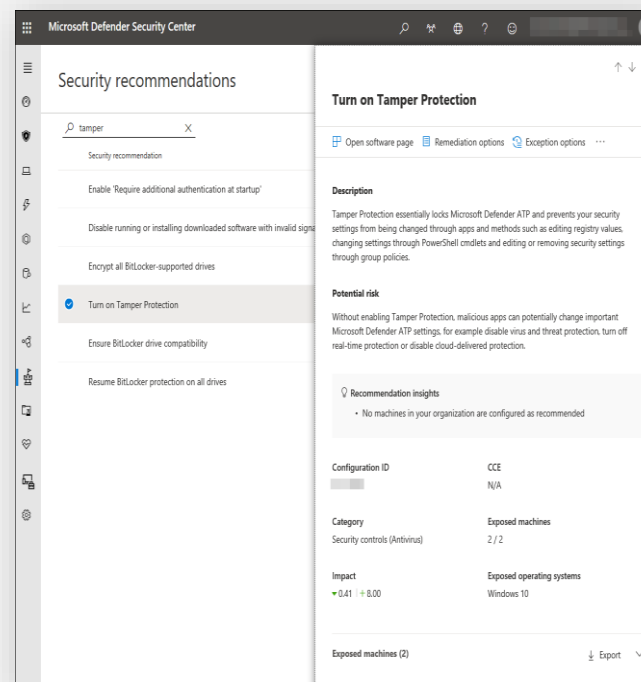
Read the [blog](#) for more details

Tamper Protection – the first step in ransomware protection

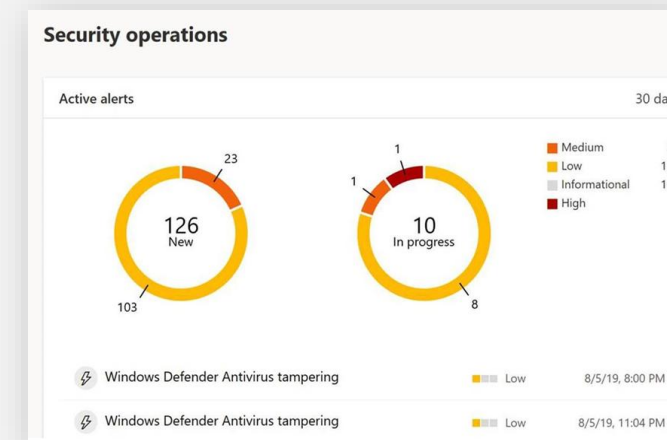
Seamless, secure and password less configuration



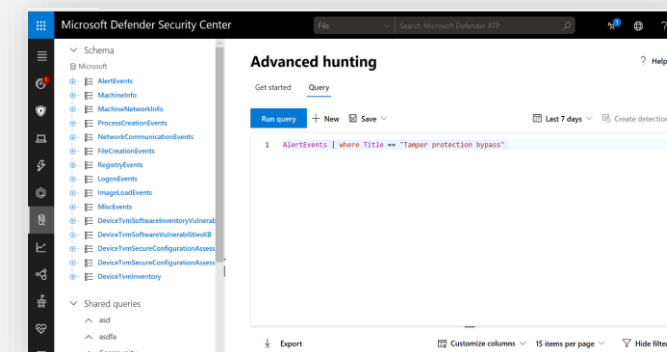
Threat & vulnerability management – Security recommendation



Tampering alert based on System Guard and EDR signals



Advanced Hunting



 Read the [blog](#) for more details

Firmware & hardware protections

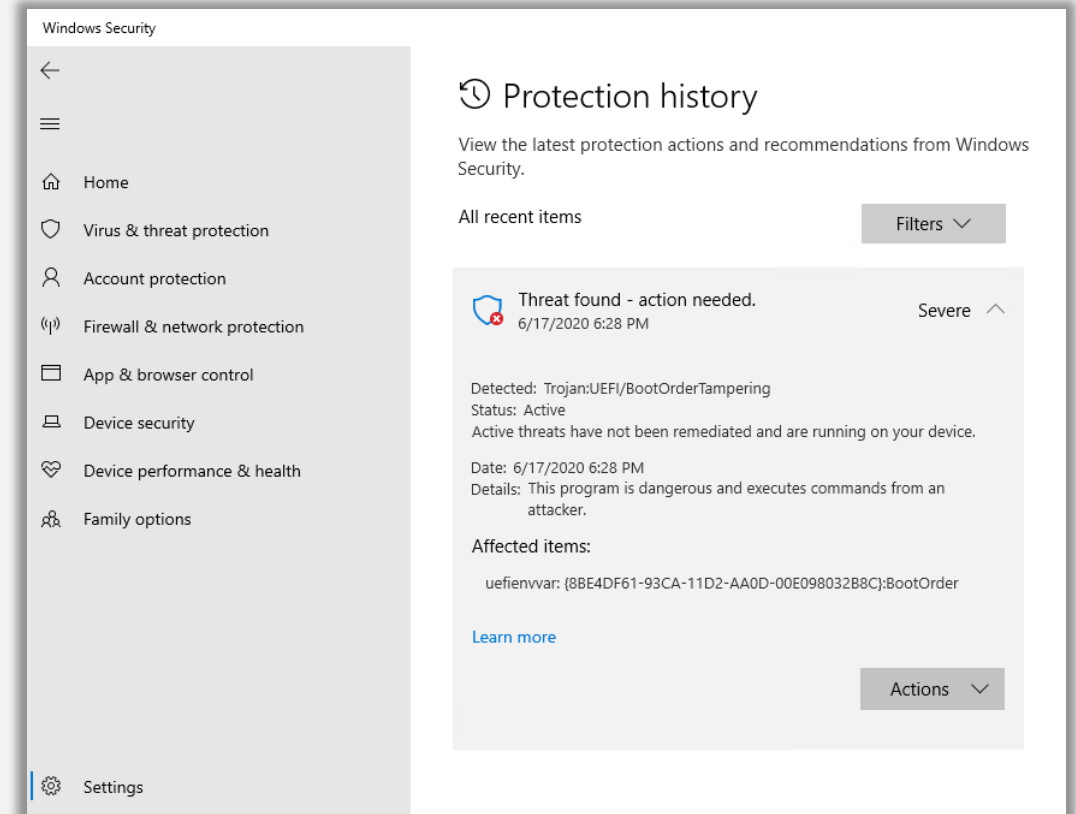
UEFI scanner reads firmware file system at runtime by interacting with the motherboard chipset, performing dynamic analysis using multiple solution components:

- UEFI anti-rootkit, which reaches the firmware through Serial Peripheral Interface (SPI)
- Full filesystem scanner, which analyzes content inside the firmware
- Detection engine, which identifies exploits and malicious behaviors

Microsoft Defender Security Center



Scanning and detection

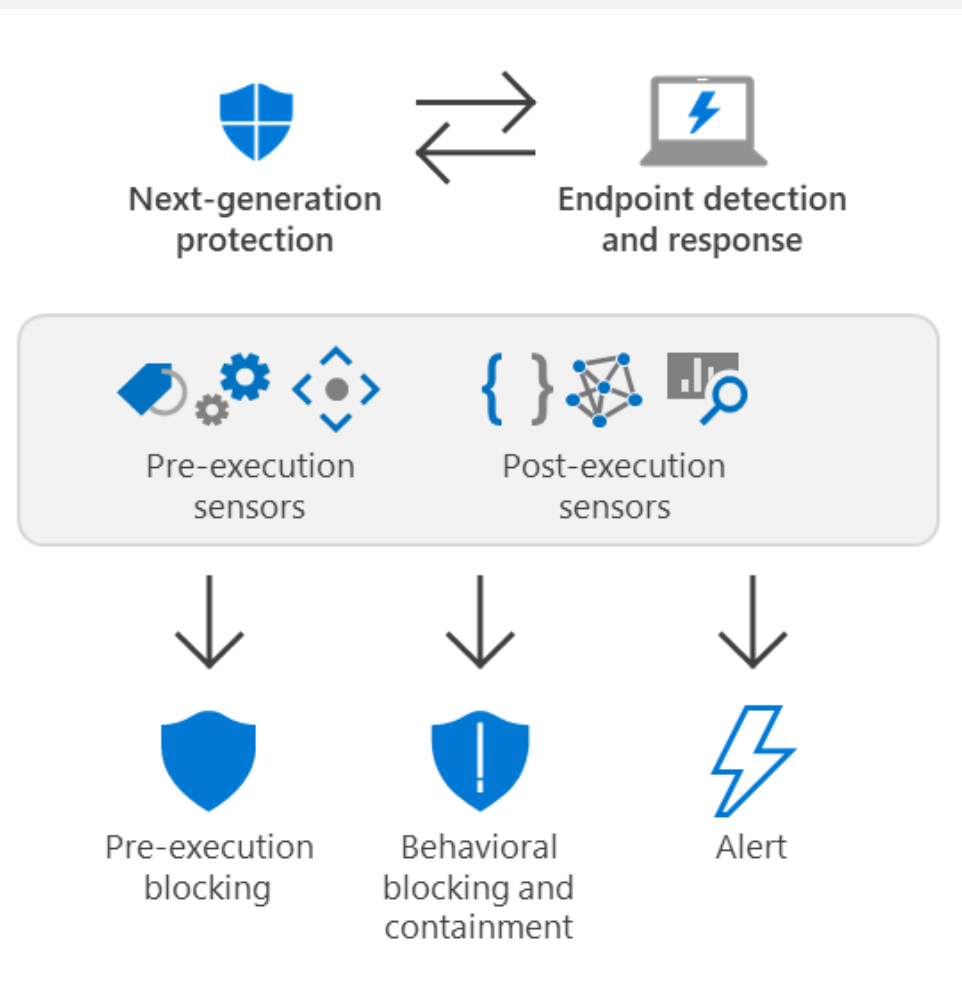


Read the [blog](#) for more details

Behavioral Blocking and Containment

- Immediately stops threat before it can progress
- Microsoft has the unique ability to scan signals across kill chains and payloads (endpoints, Office, Identity, etc.)
- Some highlights:
 - Pre and Post breach AI- and ML- based behavioral blocking and containment
 - Detect malware after first sight and block it on other endpoints within minutes (1 – 5 minutes)
 - Microsoft Defender for Endpoint provides an additional protection layer by blocking/preventing malicious behavior even if we are not the primary AV

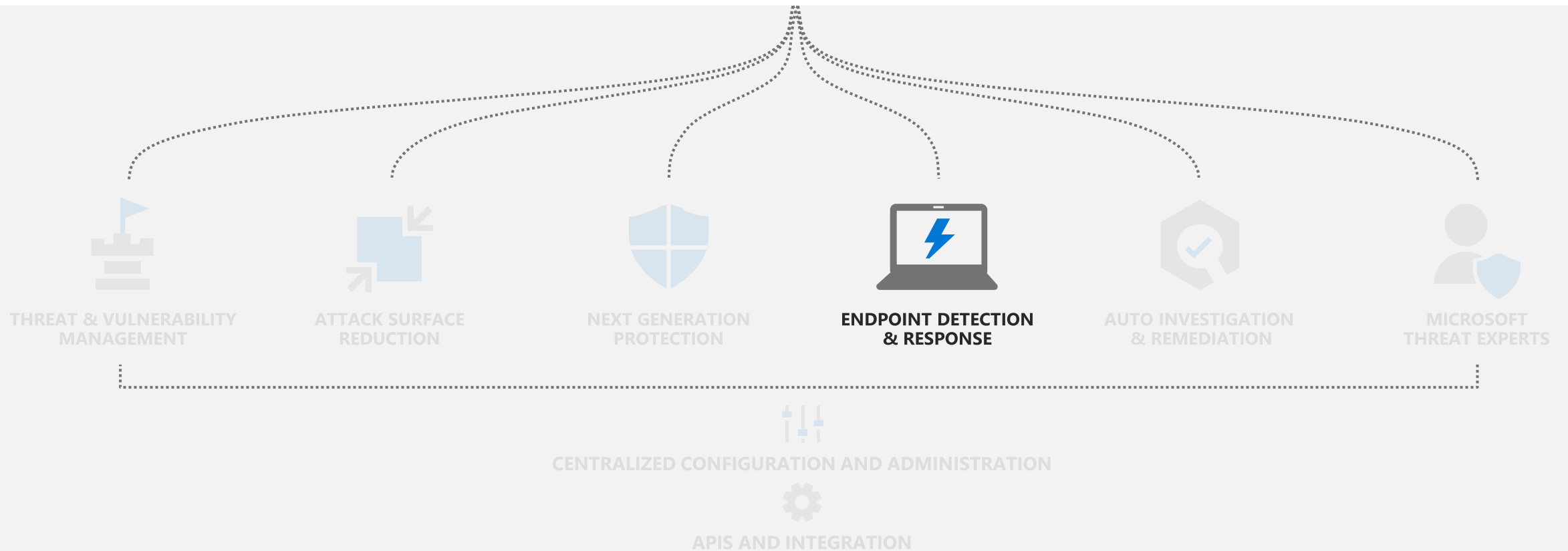
 Read the [blog](#) for more details





Next Generation Endpoint Protection

Threats are no match.



Endpoint Detection & Response

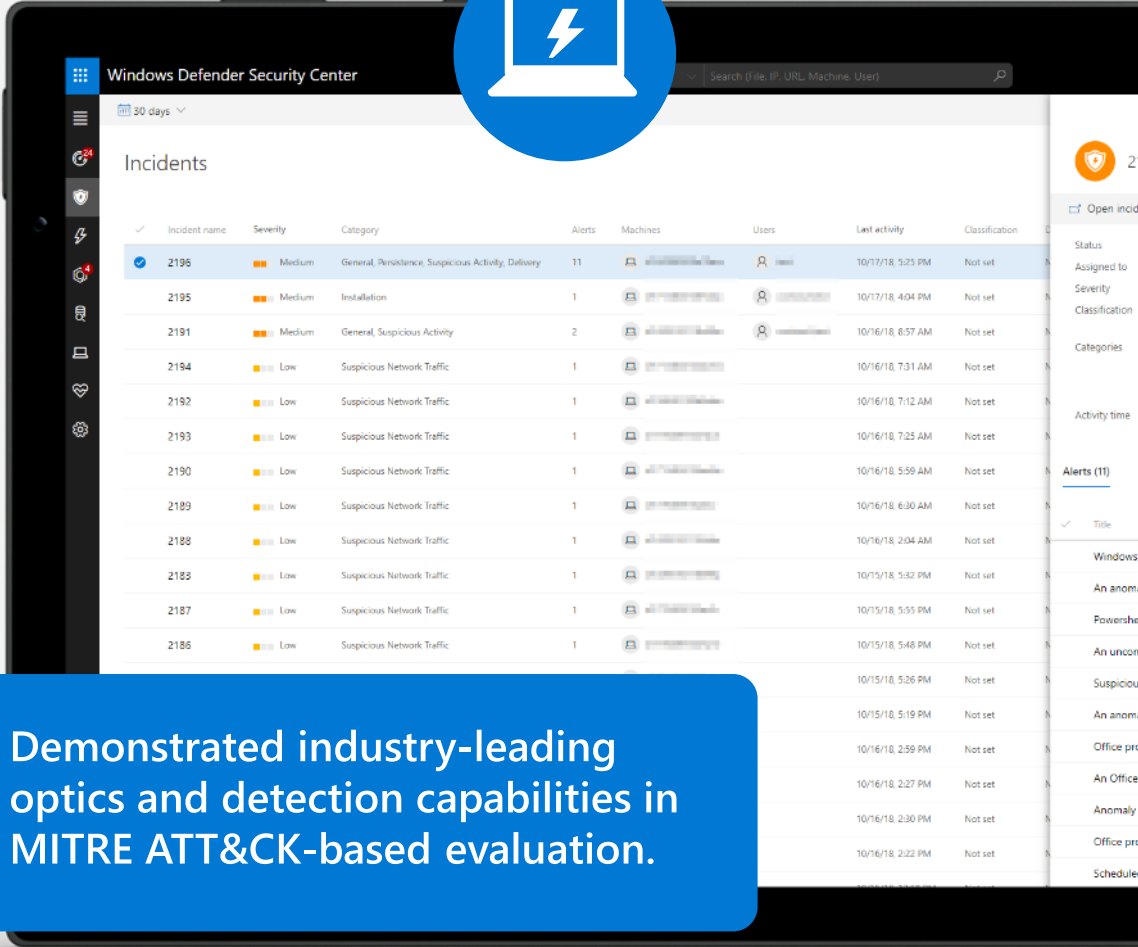
Detect and investigate advanced persistent attacks

 Correlated behavioral alerts

 Investigation & hunting over 6 months of data

 Rich set of response actions





Incident name	Severity	Category	Alerts	Machines	Users	Last activity	Classification
2196	Medium	General, Persistence, Suspicious Activity, Delivery	11			10/17/18, 5:25 PM	Not set
2195	Medium	Installation	1			10/17/18, 4:04 PM	Not set
2191	Medium	General, Suspicious Activity	2			10/16/18, 6:57 AM	Not set
2194	Low	Suspicious Network Traffic	1			10/16/18, 7:31 AM	Not set
2192	Low	Suspicious Network Traffic	1			10/16/18, 7:12 AM	Not set
2193	Low	Suspicious Network Traffic	1			10/16/18, 7:25 AM	Not set
2190	Low	Suspicious Network Traffic	1			10/16/18, 5:59 AM	Not set
2189	Low	Suspicious Network Traffic	1			10/16/18, 6:30 AM	Not set
2188	Low	Suspicious Network Traffic	1			10/16/18, 2:04 AM	Not set
2185	Low	Suspicious Network Traffic	1			10/15/18, 5:52 PM	Not set
2187	Low	Suspicious Network Traffic	1			10/15/18, 5:55 PM	Not set
2186	Low	Suspicious Network Traffic	1			10/15/18, 5:48 PM	Not set



Demonstrated industry-leading optics and detection capabilities in MITRE ATT&CK-based evaluation.

Endpoint Detection & Response



Correlated post-breach detection

Investigation experience

Incident

Advanced hunting

Response actions (+EDR blocks)

Deep file analysis

Live response

Threat analytics

Triage & Investigation

Understand what was alerted

Alert investigation experience provides detailed description, rich context, full process execution tree.

Investigate device activity

Full machine timeline to drill into activities, filter and search.

Rich supporting data & tools

Supporting profiles for files, IPs, URLs including org & world prevalence, deep analysis sandbox.

Expand scope of breach

In-context pivoting to other affected machines/users.

The image displays a collage of screenshots from a security investigation tool, illustrating various views and data points:

- control.exe file details:** Shows the file's SHA1, SHA256, MD5, and size. It includes a timeline view for the file's activity, with a custom range filter set from 8/10/2019 to 8/16/2019. The event time is 15-Aug-2019 5:39:38.755 PM on machine apt29-client3.
- control.exe created process powershell.exe:** A detailed view of the process creation event, showing the event time, action type (ProcessCreated), additional information (SuspiciousActivity), user (apt29\july.jarvis), machine (apt29-client3), and entities (sdclt.exe, control.exe, powershell.exe).
- COM hijacking alert:** An alert titled "COM hijacking" with a severity of Medium. It includes a description of the alert, a risk level of High, and recommended actions. The alert context shows it was first detected on 08.13.2019 and last active on 08.15.2019.
- Alert process tree:** A diagram showing the process execution tree, starting from sdclt.exe, which spawned control.exe, which then spawned powershell.exe, which spawned reg.exe, which spawned startup.bat, which spawned net.exe, which spawned net.exe.
- Machine details for apt29-client3:** A view showing the machine's risk level (High), exposure level (No data available), and device details (Domain: apt29.org, OS: Windows 10 x64, Version: 1903, Build: 18362, Health state: Inactive).
- Timeline view:** A timeline view showing events on the machine apt29-client3. It includes a search bar, a custom range filter (8/8/2019 to 8/15/2019), and a list of events with their times and descriptions. The events include: "COM hijacking", "reg.exe set registry value for key 'HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run'", "powershell.exe created process reg.exe", "powershell.exe ran Powershell command: 'reg.exe'", "SearchIndexer.exe created process SearchProtocolHost.exe", "An uncommon file was created and added to startup folder", "powershell.exe created file startup.bat", and "powershell.exe ran Powershell command: 'Set-Content'".

Incidents

Narrate the end-to-end attack story

Reconstructing the story

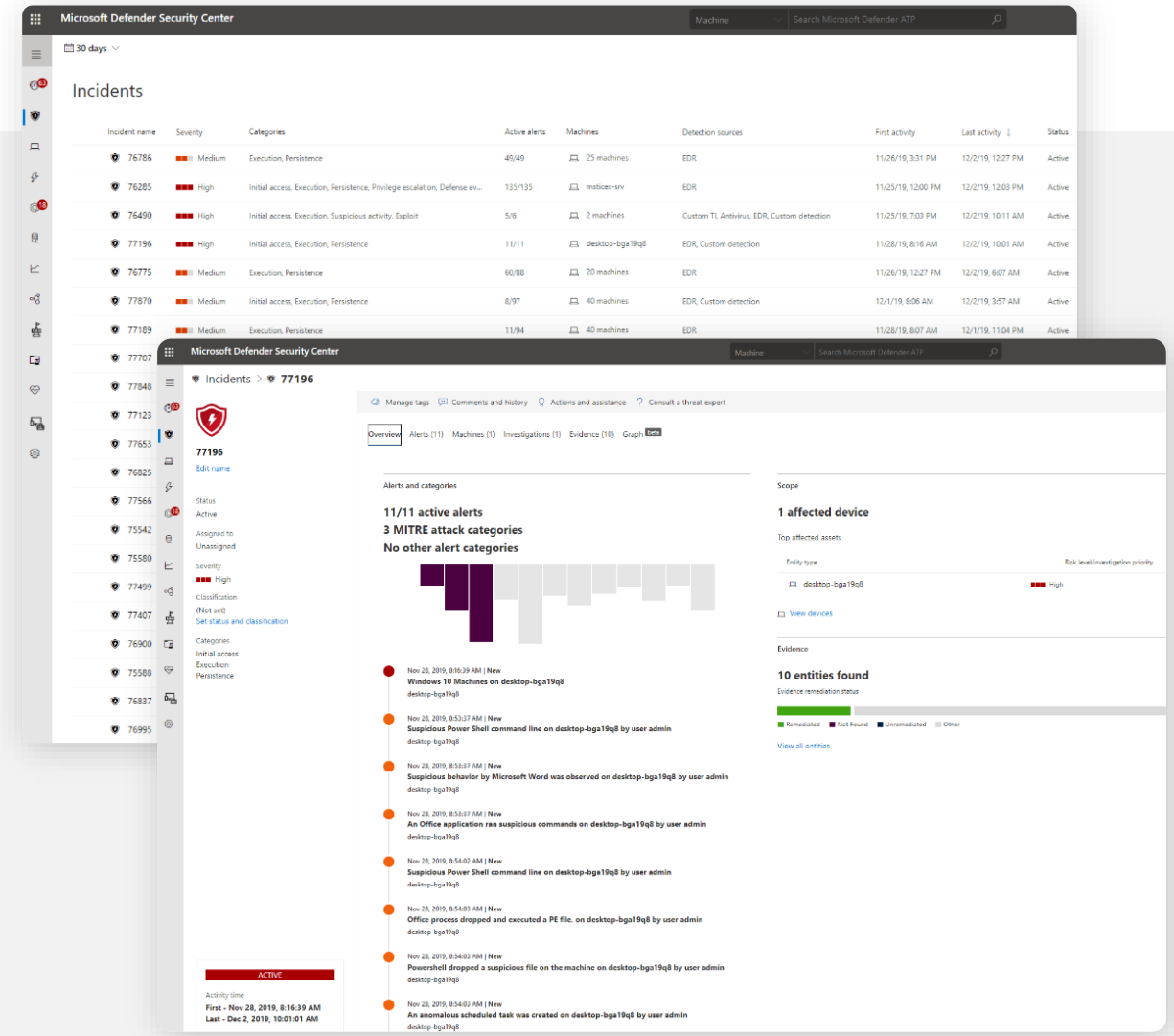
The broader attack story is better described when relevant alerts and related entities are brought together.

Incident scope

Analysts receive better perspective on the purview of complex threats containing multiple entities.

Higher fidelity, lower noise

Effectively reduces the load and effort required to investigate and respond to attacks.



[Announcement blog](#)

Advanced hunting with custom detection and custom response

The screenshot displays the Microsoft Defender Security Center interface, specifically the 'Advanced hunting' section. The left sidebar shows a navigation menu with categories like Schema, Shared queries, and Test. The main area is titled 'Advanced hunting' and shows a query for 'PowerShell downloads'. The query is a KQL (Kusto Query Language) script that filters for PowerShell execution events that could involve a download, based on file name and process command line. The results are displayed in a table with columns: EventTime, ComputerName, InitiatingProcessFileName, and FileName. The table shows 15 items per page, with 1-15 of 100 results visible. The right sidebar contains filters for ComputerName, InitiatingProcessFileName, FileName, and ProcessCommandLine.

Microsoft Defender Security Center

Machine | Search Microsoft Defender ATP | AutoIR@wcdtestprd.onmicr...

Advanced hunting

Get started | PowerShell downloads

Run query | + New | Save | Last 7 days | Create detection rule

```
1 // Finds PowerShell execution events that could involve a download.
2 ProcessCreationEvents
3 | where EventTime > ago(7d)
4 | where FileName in ("powershell.exe", "POWERSHELL.EXE", "powershell_ise.exe", "POWERSHELL_ISE.EXE")
5 | where ProcessCommandLine has "Net.WebClient"
6 |   or ProcessCommandLine has "DownloadFile"
7 |   or ProcessCommandLine has "Invoke-WebRequest"
8 |   or ProcessCommandLine has "Invoke-Shellcode"
9 |   or ProcessCommandLine contains "http:"
10 | project EventTime, ComputerName, InitiatingProcessFileName, FileName, ProcessCommandLine
11 | top 100 by EventTime
12
```

Export | Customize columns | 15 items per page | 1-15 of 100 | Hide filters

EventTime	ComputerName	InitiatingProcessFileName	FileName
12/2/2019 12:02:32 PM	msticex-srv.msticex.net	cmd.exe	powershell.exe
12/2/2019 12:01:32 PM	msticex-srv.msticex.net	cmd.exe	powershell.exe
12/2/2019 12:01:32 PM	msticex-srv.msticex.net	cmd.exe	powershell.exe
12/2/2019 12:01:31 PM	msticex-srv.msticex.net	cmd.exe	powershell.exe
12/2/2019 2:51:10 AM	tk5-3wp03r0809.cfdev.nttest.microsoft.com	cmd.exe	powershell.exe
12/2/2019 2:47:26 AM	tk5-3wp03r0813.cfdev.nttest.microsoft.com	cmd.exe	powershell.exe
12/1/2019 19:26:27 PM	tk5-3wp03r0801.cfdev.nttest.microsoft.com	cmd.exe	powershell.exe
12/1/2019 18:35:42 PM	tk5-3wp03r0801.cfdev.nttest.microsoft.com	cmd.exe	powershell.exe
12/1/2019 18:35:42 PM	tk5-3wp03r0809.cfdev.nttest.microsoft.com	cmd.exe	powershell.exe
12/1/2019 18:35:42 PM	tk5-3wp03r0813.cfdev.nttest.microsoft.com	cmd.exe	powershell.exe

Filters

ComputerName

Search

InitiatingProcessFileName

Search

FileName

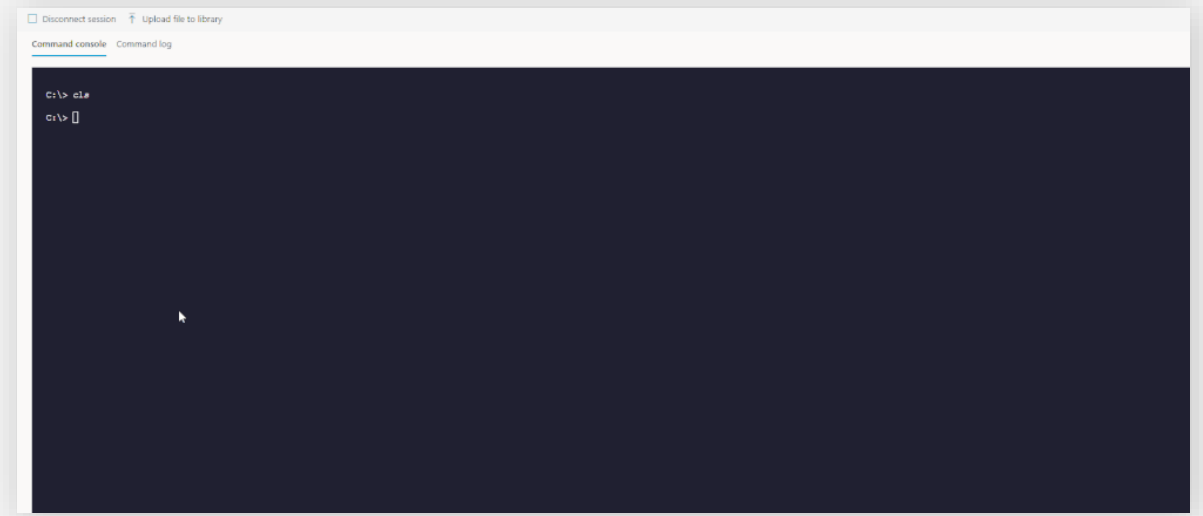
Search

ProcessCommandLine

Search

Live Response

- Real-time live connection to a remote system
- Leverage Microsoft Defender for Endpoint Auto IR library (memory dump, MFT analysis, raw filesystem access, etc.)
 - Extended remediation command + easy undo
- Full audit
- Extendable (write your own command, build your own tool)
- RBAC+ Permissions
- Git-Repo (share your tools)



Threat Analytics

Delivering insight on major threats to your organization

Threat to posture view

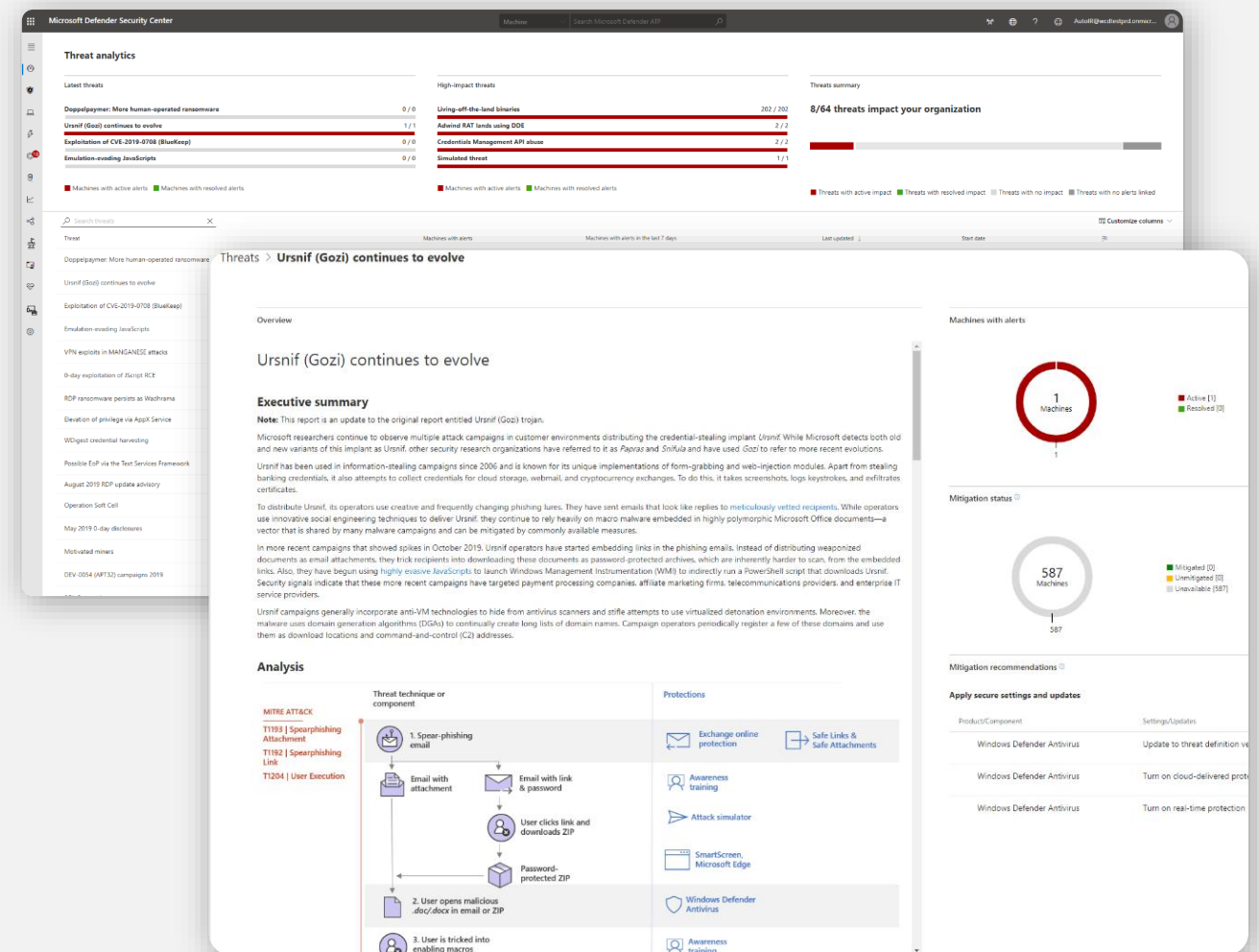
See how you score against significant and emerging campaigns with interactive reports.

Identify unprotected systems

Get real-time insights to assess the impact of the threat on your environment.

Get guidance

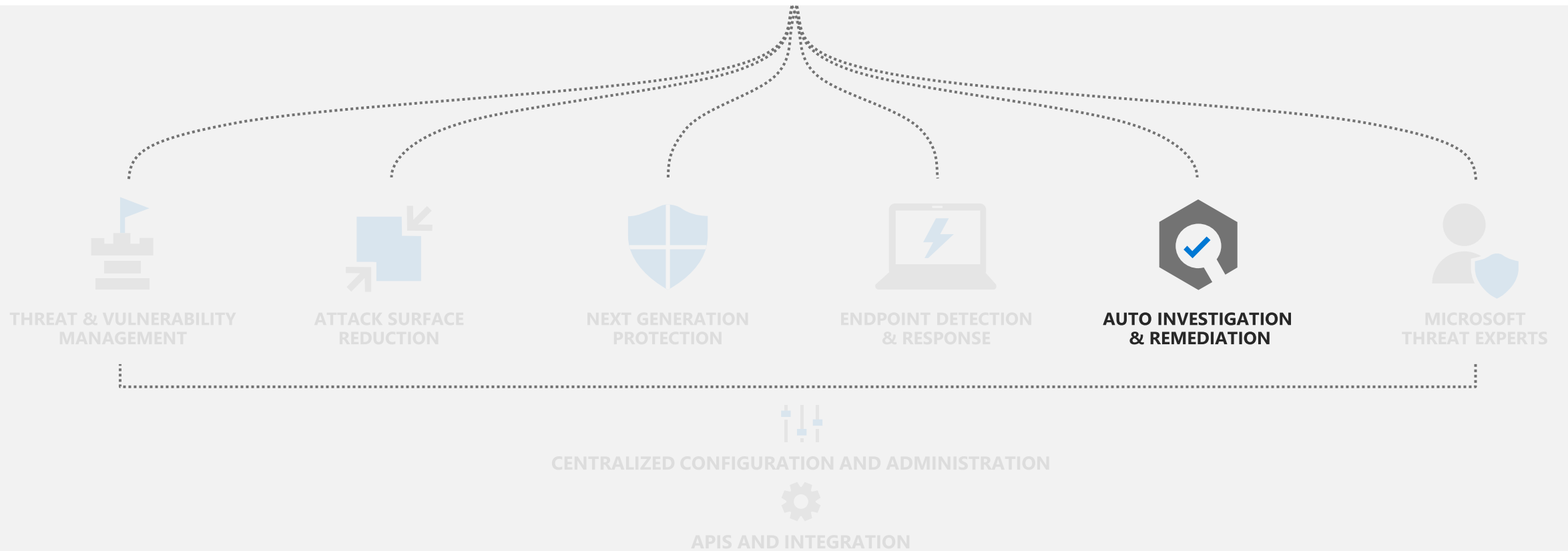
Provides recommended actions to increase security resilience, to prevention, or contain the threat.





Next Generation Endpoint Protection

Threats are no match.



Auto Investigation & Remediation

Automatically investigates alerts and remediates complex threats in minutes



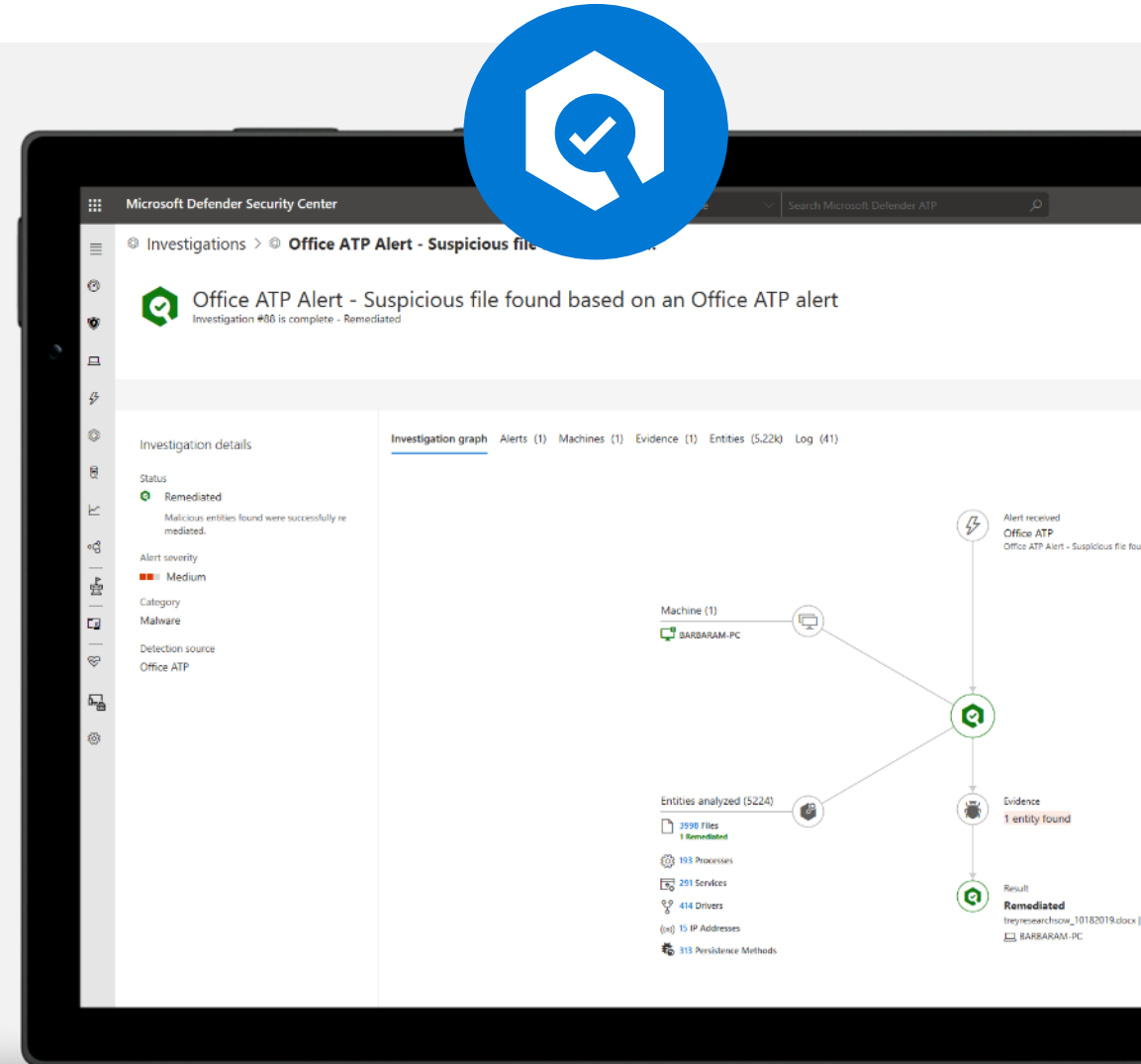
Mimics the ideal steps analysts would take



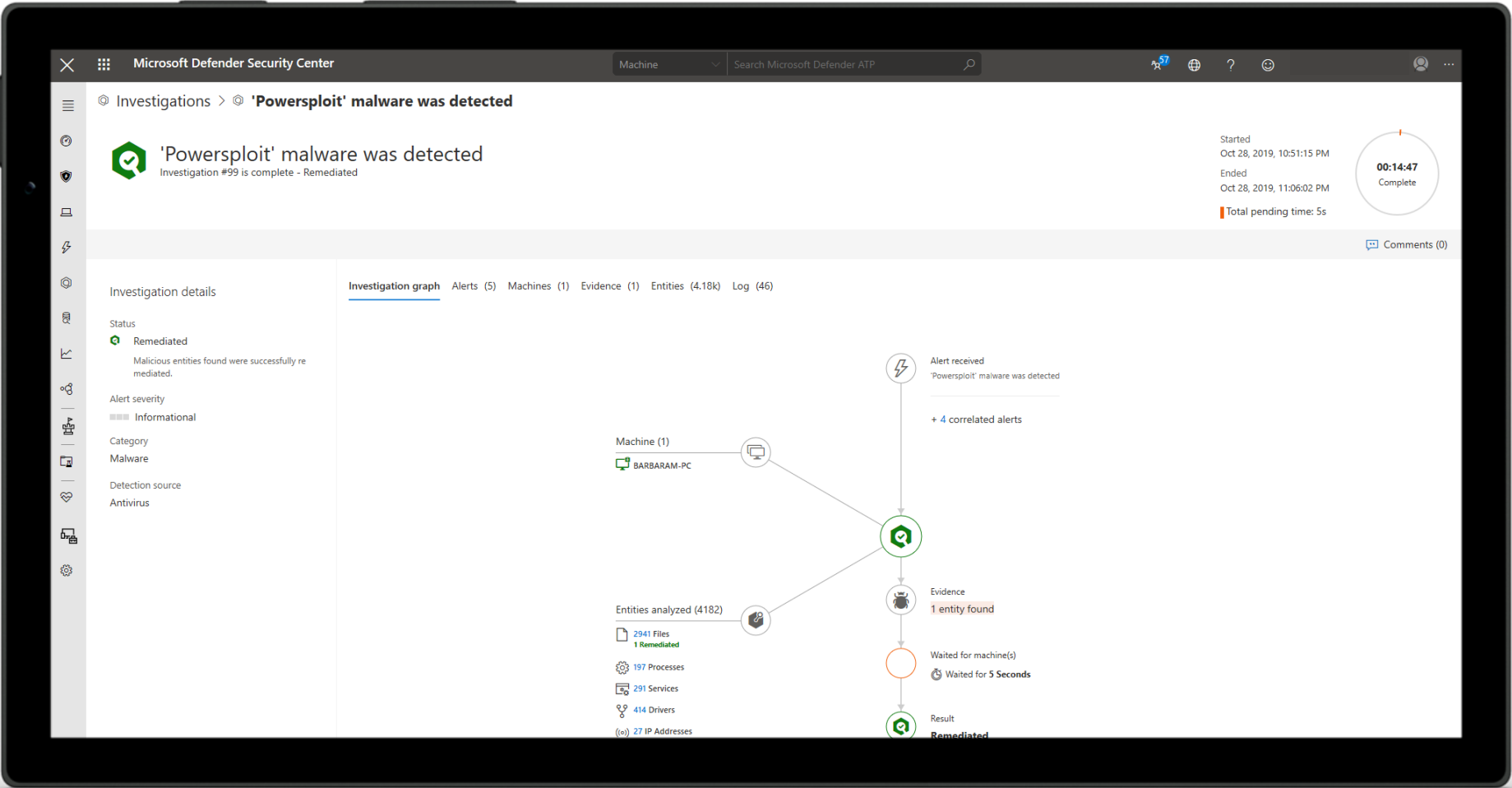
Tackles file or memory-based attacks



Works 24x7, with unlimited capacity



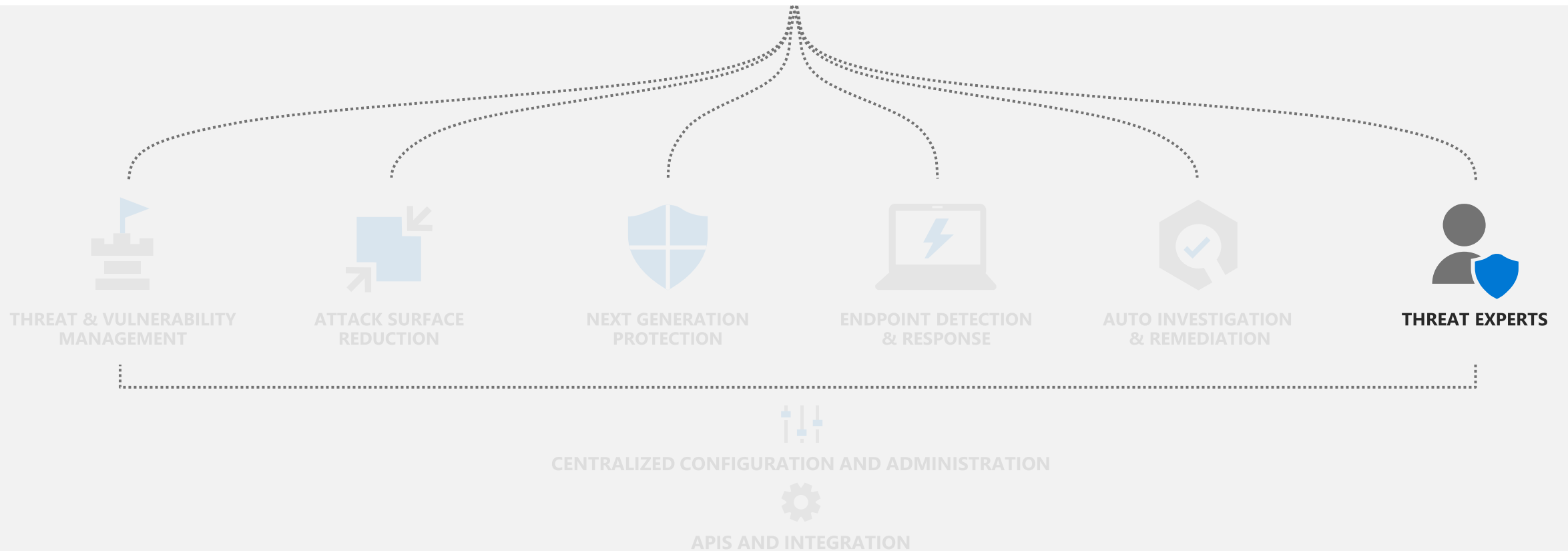
Investigation graph





Next Generation Endpoint Protection

Threats are no match.



Threat Experts

Bring deep knowledge and proactive threat hunting to your SOC



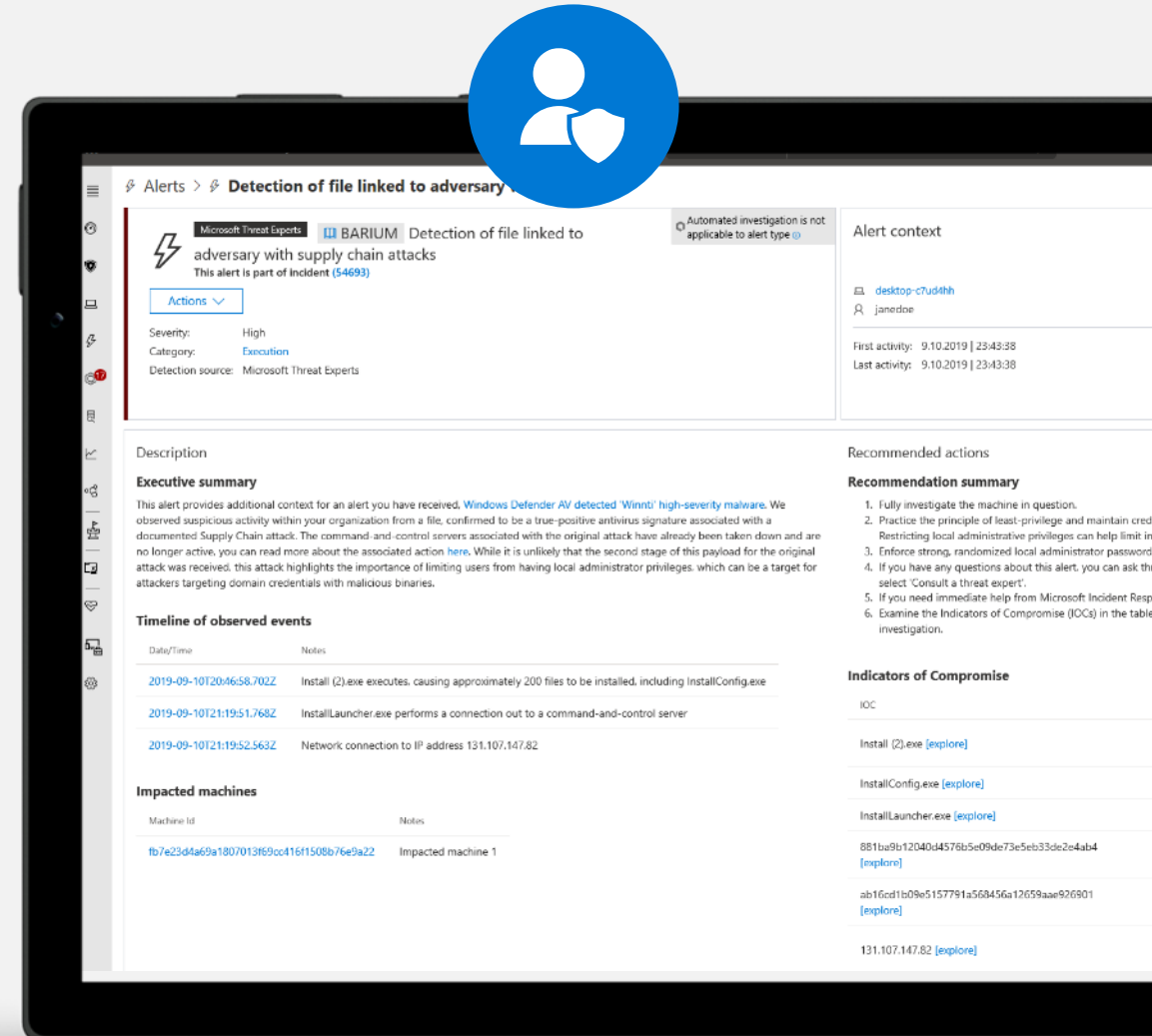
Expert level threat monitoring and analysis



Environment-specific context via alerts



Direct access to world-class hunters



Threat Experts

An additional layer of oversight and analysis to help ensure that threats don't get missed

Targeted attack notifications

Threat hunters have your back.

Microsoft Threat Experts proactively hunt to spot anomalies or known malicious behavior in your unique environment.

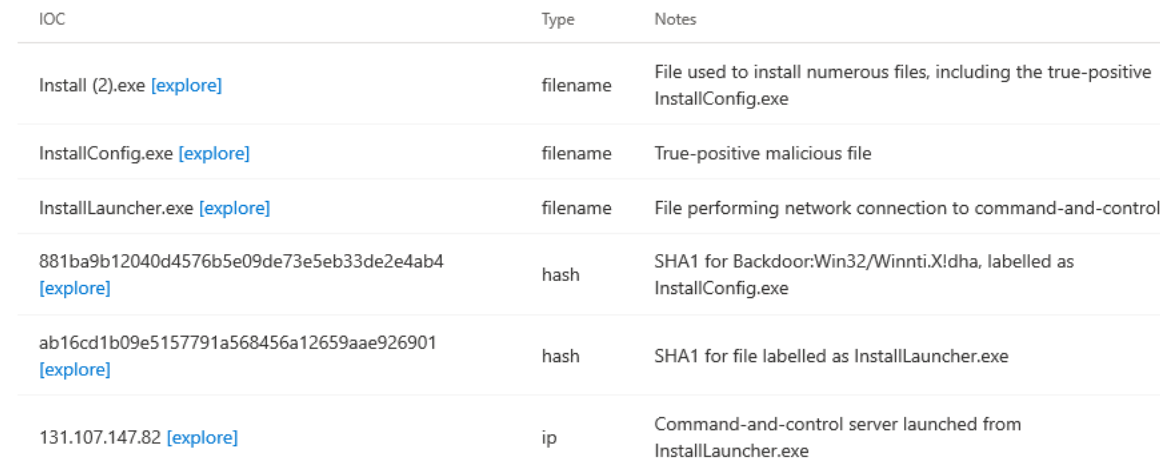
Experts on demand

World-class expertise at your fingertips.

Got questions about alert, malware, or threat context? Ask a seasoned Microsoft Threat Expert.

The screenshot displays the Microsoft Defender Security Center interface. The top section shows an alert titled "Detection of file linked to adversary with supp..." with a severity of High and a category of Execution. The alert context includes details about the file and the user. Below the alert, the "Description" tab is active, showing a "Software Supply Chain Attack" dashboard. The dashboard includes a "10 active alerts" bar chart, a "Related evidence" section with 6 machines, 6 users, 1 malware, and 8 files, and a table of alerts. The table lists various suspicious activities such as "Software Supply Chain Attack", "Suspicious PowerShell", "Suspect scheduled task", "Active credential theft tool", "Suspicious user behavior", and "Compromised Account". The bottom right section shows the "Alert description" and "Recommended actions".

Title	Severity	Investigation state	Category	Machine	User
Software Supply Chain Attack	High	N/A	Bilbao	cont-jonathanw	JW
Suspicious PowerShell	Medium	Running	Suspicious Activity	cont-jonathanw	JW
Suspect scheduled task	Medium	Running	Persistence	cont-jonathanw	JW
Active credential theft tool	High	Running	Credential Theft	cont-jonathanw	JW
Suspicious user behavior	Low	Running	Compromised Account	cont-jayhandse	JW
Suspicious PowerShell	Medium	Running	Suspicious Activity	cont-jayhandse	JW
Suspect scheduled task	Medium	Running	Persistence	cont-jayhandse	JW
Active credential theft tool	High	Running	Credential Theft	cont-jayhandse	JW
Suspicious user behavior	Low	Running	Compromised Account	cont-jayhandse	JW



Microsoft Threat Experts - Trial

Your Experts on Demand trial version expires in 41 days from your Microsoft Threat Experts enrolment. Contact your Microsoft representative to get a full subscription.

Learn more about [Microsoft Threat Experts – Experts on Demand](#)



Consult a threat expert

Get Microsoft Threat Experts advice and insights about suspicious activities in your organization.

Ensure that the portal page for the alert or machine in question is in view while providing information for this inquiry.

Note: This and other relevant information will be shared with Microsoft Threat Experts to enable the best response to your inquiry.

Inquiry topic

https://securitycenter.windows.com/alert/da637073841040265613_-882982118

Thank you for sending this Threat Expert alert. Can you help us investigate this threat further including whether you think we were targeted, and whether this and other machines in our company were compromised?

Email ✱

Enter the email address you'd like Microsoft Threat Experts to send their reply

Analyst@contoso.com

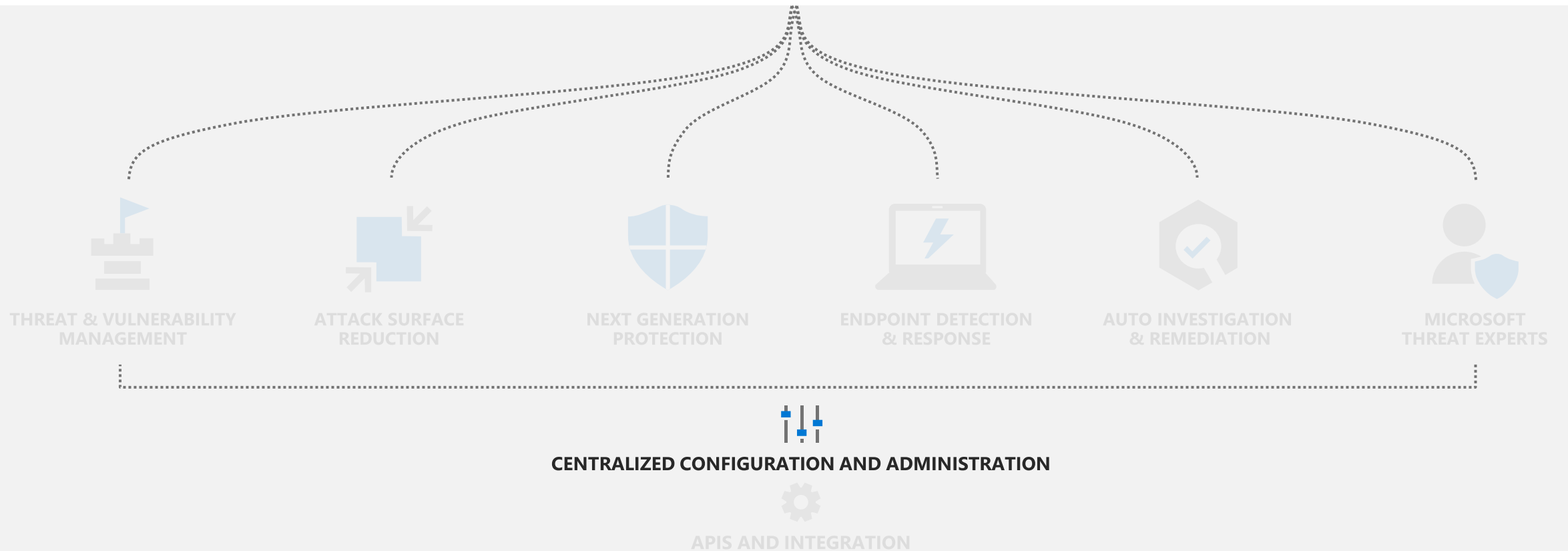
Submit

[Privacy statement.](#)



Next Generation Endpoint Protection

Threats are no match.



Historical roles & friction



Security Team

- Responsible for security monitoring and reducing risk
- Analyze threats, security incidents, exposure and identify mitigations
- Define security policies
- Priority is on quick remediation on impacted devices/users



IT Team

- Responsible for policy configuration including security policies
- Analyzes change impact and stages rollout of global policies
- Priority is a stable IT environment and low costs

Security Management

Assess, configure and respond to changes in your environment



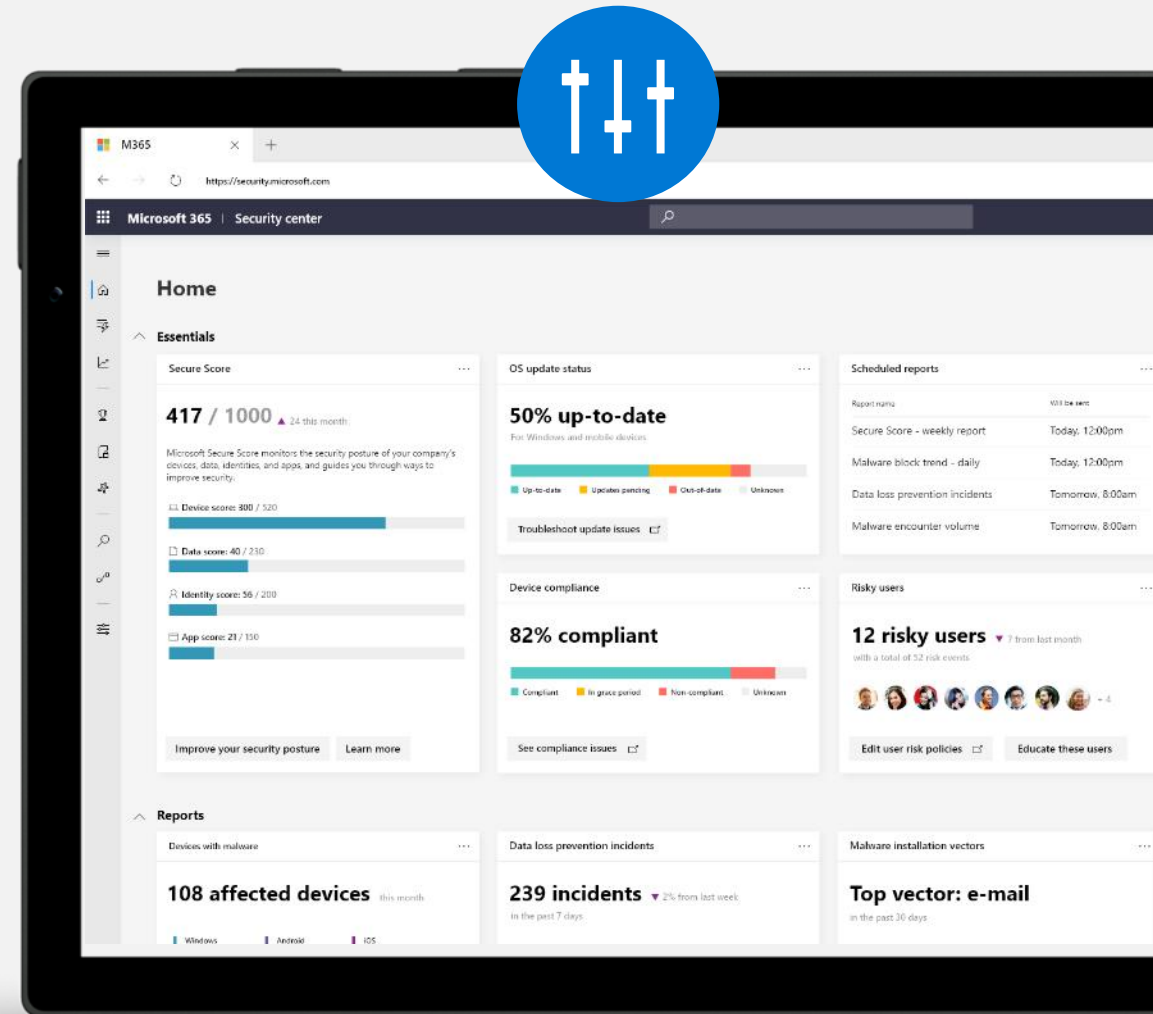
Centrally assess & configure your security



Variety of reports and dashboards for detailed monitoring and visibility



Seamless integration between policy assessment and policy enforcement



Endpoint Security Management



All
devices



Sec Admin
experiences



Security
baselines



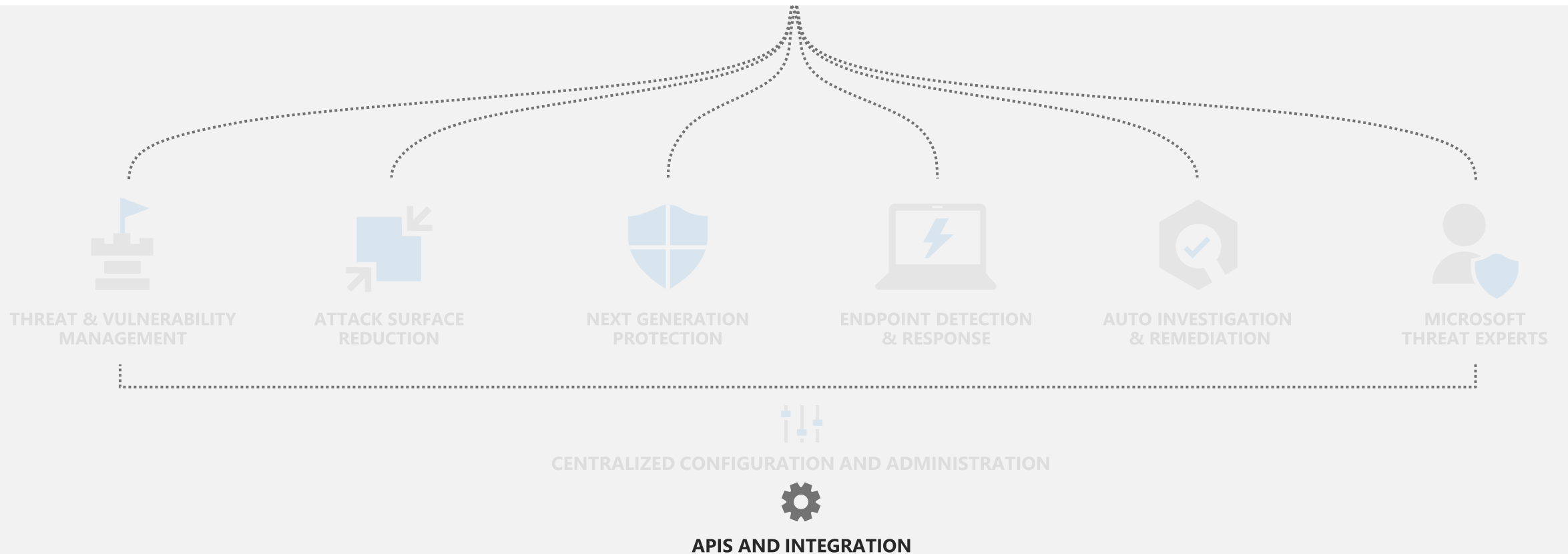
Security
tasks

Target security policy to any device across Windows, Mac, Linux, Android, or iOS

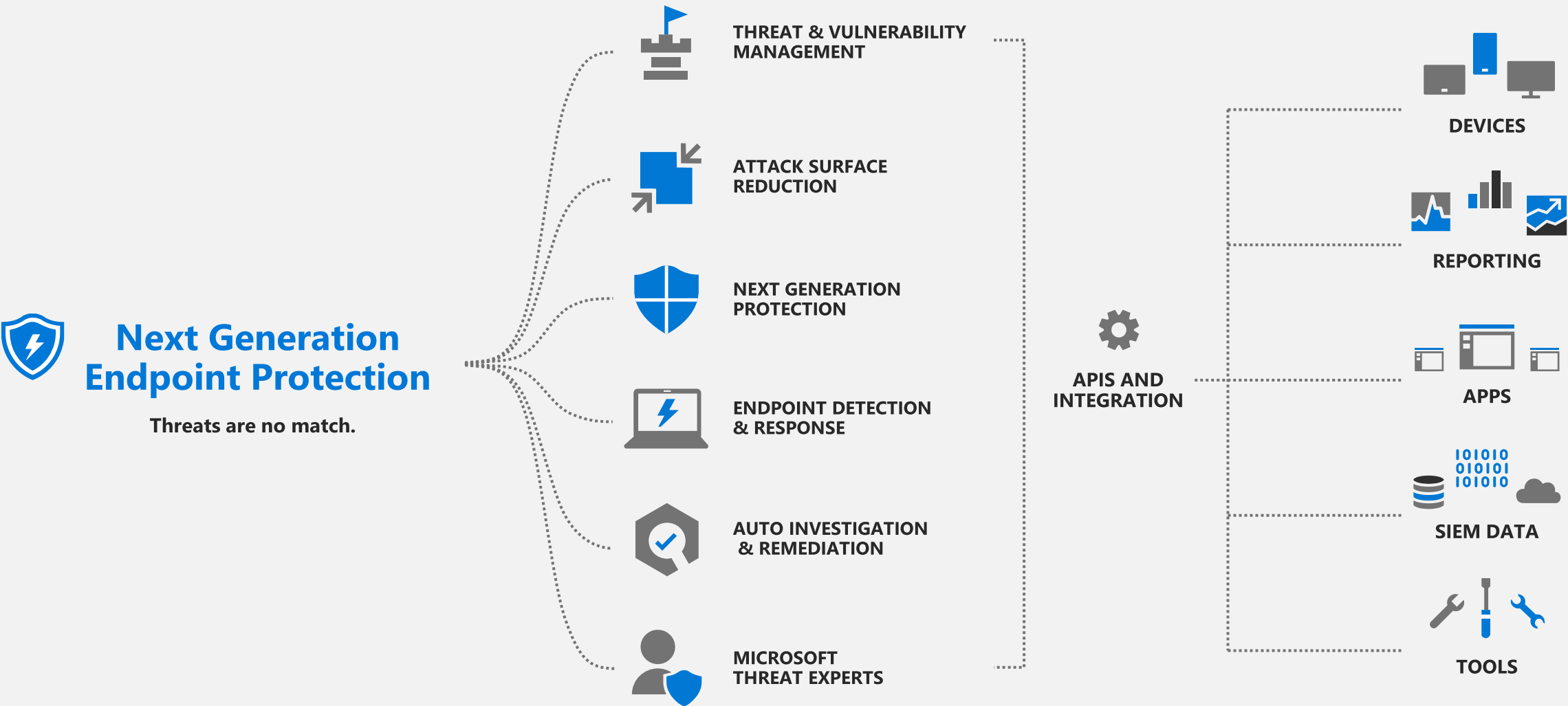


Next Generation Endpoint Protection

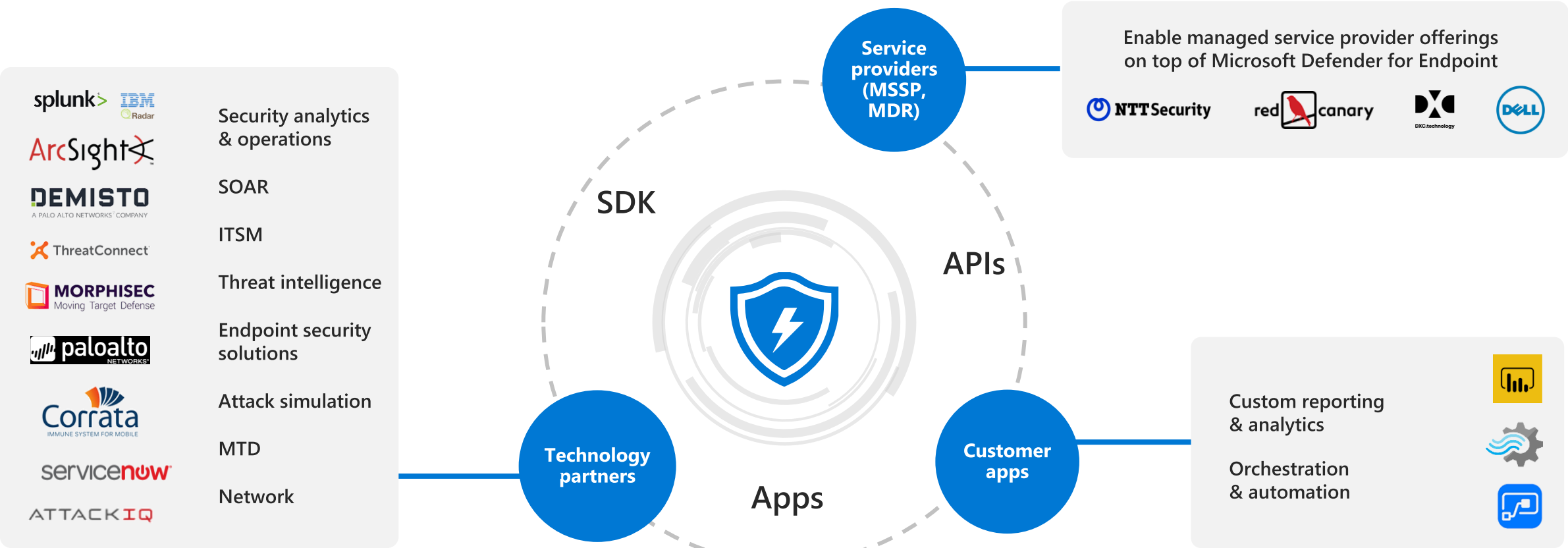
Threats are no match.



Connecting with the platform



NG Endpoint through ecosystem & API



- + Query API
- + Streaming API
- + Actions API

- + Threat intel API, Vulnerability API
- + Application connectors (PBI, Flow, SNOW)
- + Microsoft Security Graph connector

- + AAD authentication & authorization
- + RBAC controls

- + Developer kit
- + Partner integration kit
- + Developer License